

Guided Flow Matching for Forward and Inverse PDE Problems with Sparse Observations: Algorithm and Theory

Xifeng Zhang

*School of Mathematical Science
Capital Normal University
Beijing, 100048, China*

<CNXIFENG9819@163.COM>

Jin Zhao*

*Academy for Multidisciplinary Studies
Capital Normal University
Beijing, 100048, China*

<ZJIN@CNU.EDU.CN>

Abstract

We propose FM4PDE, an endpoint-guided flow-matching framework for forward, inverse, and sparse-reconstruction problems governed by partial differential equations. The model learns a joint generative prior over coefficients or initial states and solution or terminal fields, and it imposes sparse observations and physical constraints only at inference time. A central design choice is to evaluate observation and PDE-residual losses on decoded endpoint predictions in physical units, rather than on noisy intermediate flow states. We derive an endpoint-posterior interpretation of the guidance rule and analyze the implemented finite-step sampler under explicit finite-horizon localization, local smoothness, local PL-type geometry, and endpoint-transport defect assumptions. The deterministic, stochastic, and hybrid bounds are conditional and floor-aware: they quantify descent or contraction only above discretization, approximation, proposal, and transport-error floors. The experimental protocol is designed to compare FM4PDE with neural-operator, sparse-reconstruction, physics-constrained optimization, and data-assimilation baselines across static and time-dependent PDE families.

Keywords: flow matching, partial differential equations, sparse observations, PDE constraints, error analysis

1 Introduction

Partial differential equations (PDEs) are fundamental mathematical models for physical, biological, and engineering systems. In computational science, one often needs to solve forward problems, where the governing coefficient or initial state is known and the solution field is sought, and inverse problems, where unknown coefficients, sources, or initial states must be inferred from observations of the solution. In many applications, however, observations are sparse, irregular, noisy, or available only at a few time snapshots because sensors are expensive, spatial access is limited, or repeated full-field simulation is impractical. Sparse-observation PDE reconstruction is therefore underdetermined unless one combines data-driven priors, physical constraints, and task-specific conditioning.

Classical numerical solvers, PDE-constrained optimization, and variational data assimilation provide strong physical fidelity, but they typically require solving a new discretized

*. Corresponding author.

optimization, adjoint, or assimilation problem for each observation instance (Hinze et al., 2008; Le Dimet and Talagrand, 1986). Physics-informed neural networks (PINNs) embed PDE residuals and boundary or initial conditions in the loss and can be applied to forward and inverse problems with scattered measurements (Raissi et al., 2019). Bayesian and physics-constrained variants further address noisy data and uncertainty quantification in sparse-field reconstruction (Yang et al., 2021; Sun and Wang, 2020). These methods are important comparators for sparse inverse problems, but they usually require per-instance optimization or posterior inference at test time.

Neural operators provide a complementary amortized viewpoint. Fourier Neural Operators (FNO), DeepONet, and more general neural-operator frameworks learn maps between function spaces from paired simulation data and can be highly efficient once trained (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023). Vanilla operator learners are most naturally used when full input-output fields are available, but a growing literature has adapted operator learning and deep reconstruction architectures to sparse sensors, irregular observations, and inverse settings. Examples include Voronoi-tessellation-assisted sparse field reconstruction (Fukami et al., 2021), RecFNO for resolution-invariant flow and heat reconstruction from sparse observations (Zhao et al., 2024), the attention-based Senseiver for sparse field reconstruction (Santos et al., 2023), VIVID-style deep data assimilation with sparse and time-varying sensors (Cheng et al., 2024), and invertible Fourier neural operators for joint forward and inverse problems (Long et al., 2025). Hence the contribution of FM4PDE should not be read as claiming that sparse-observation PDE reconstruction is unaddressed. Rather, the goal is to develop a single generative-prior framework that supports forward, inverse, and sparse-reconstruction tasks through conditional sampling.

Generative models are attractive for this purpose because they can represent multi-modal priors over coefficients, initial states, and solution fields. Diffusion and score-based models learn distributions over data by reversing a noising process and can be conditioned by observations or additional energy terms (Ho et al., 2020; Song et al., 2021; Ho and Salimans, 2022). In PDE settings, generative methods such as DiffusionPDE and physically conditioned score-based models use learned priors together with partial observations or physical constraints for conditional generation (Huang et al., 2024; Jacobsen et al., 2025). Flow matching learns a velocity field that transports a simple source distribution to the data distribution along a prescribed probability path (Lipman et al., 2023, 2024; Liu et al., 2023). Compared with diffusion samplers, flow-matching samplers can use deterministic ODE integration or a smaller number of stochastic reinjections, making them attractive for fast conditional reconstruction.

A central issue, however, is how to apply observation and PDE guidance during flow-matching sampling. The raw intermediate state at flow time t is generally a noisy interpolation state, not a physically meaningful PDE field. Directly applying a finite-difference PDE residual to such a state can introduce numerical artifacts. FM4PDE therefore uses endpoint-guided losses: the sparse observation loss and the discretized PDE residual are evaluated on an endpoint prediction decoded back to physical units, rather than on the raw noisy flow state. This distinction is important for time-dependent systems, where physical time is denoted by τ and is separate from the flow-matching time t .

1.1 Related Works

Physics-informed learning, neural operators, and sparse reconstruction. PINNs and their Bayesian or physics-constrained extensions provide flexible tools for forward and inverse PDE problems with scattered or noisy measurements (Raissi et al., 2019; Yang et al., 2021; Sun and Wang, 2020). Neural operators, including FNO and DeepONet, learn amortized solution operators from simulation datasets (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023). Sparse-field reconstruction and data-assimilation methods explicitly address partial observations, including Voronoi-based reconstructions, sparse-observation neural operators, attention-based irregular-sensor architectures, and deep variational assimilation methods (Fukami et al., 2021; Zhao et al., 2024; Santos et al., 2023; Cheng et al., 2024; Long et al., 2025).

Generative models for PDE solving. DiffusionPDE learns a joint distribution of coefficients and solutions and performs conditional generation for PDE tasks (Huang et al., 2024). Other physics-conditioned generative approaches use score guidance, physically consistent conditioning, ControlNet-style conditioning, or residual-based sampling (Zhang et al., 2023; Jacobsen et al., 2025). Recent physics-constrained flow-matching models also show that constraints can be incorporated during generation (Utkarsh et al., 2025; Baldan et al., 2025). FM4PDE differs in its emphasis on paired coefficient–solution or initial–terminal datasets, endpoint-based observation and PDE guidance at inference time, and conditional finite-step analyses of deterministic, stochastic, and hybrid guided samplers.

1.2 Contributions

We propose FM4PDE (Flow Matching for PDE-Solving), a guided flow-matching framework for sparse-observation PDE reconstruction. The main contributions are as follows.

- (i) **Endpoint-guided flow-matching framework.** FM4PDE learns a joint generative prior over PDE coefficients or initial states and corresponding solution or terminal fields. During inference, sparse observation losses and PDE residual losses are evaluated on endpoint predictions decoded back to physical units. This formulation avoids interpreting raw noisy flow states as physical PDE fields.
- (ii) **Conditional deterministic finite-step analysis.** We analyze the implemented globally clipped finite-step update under finite-horizon proposal growth, local endpoint smoothness, and local PL-type geometry on a high-probability localization region. The resulting estimate gives phase-wise descent or contraction with explicit proposal, endpoint-transport, and discretization defect terms, rather than an unconditional global convergence guarantee.
- (iii) **Conditional stochastic and hybrid analyses.** For stochastic endpoint reinjection, we provide a unified defect-based expected bound and a complementary endpoint-space decomposition separating endpoint bias, Jacobian fluctuation, noise fluctuation, and curvature effects. The hybrid deterministic–stochastic result composes phase-wise bounds and is explicitly floor-aware; it does not assert that hybrid sampling uniformly improves over pure stochastic sampling.

- (iv) **Mechanism-level constant-guidance example.** We include a one-dimensional linear construction showing how repeated terminal-noise reinjection with constant guidance can retain a positive variance floor. This example motivates adaptive guidance, but it is not claimed to be a universal lower bound for all constant-guidance algorithms, all grids, or all PDE losses.
- (v) **Reproducible experimental specification.** We revise the experimental section to specify PDE forms, residuals, boundary and initial-condition penalties, normalization, neural architectures, training details, guidance hyperparameters, baseline categories, observation-only guidance controls, sensor/noise sweeps, mean-standard-deviation reporting, and negative-case discussion.

1.3 Paper Organization

Section 2 introduces the guided flow-matching formulation and the endpoint-based losses. Section 3 presents the conditional deterministic, stochastic, and hybrid analyses, including the role of clipping and additive floors. Section 4 describes the revised experimental protocol. Section 5 discusses the interpretation and limitations of the theory and experiments.

2 Methodology

FM4PDE learns an unconditional joint prior over PDE input-output pairs and incorporates sparse observations and physical constraints only at inference time. We distinguish three time or state variables throughout. The variable $t \in [0, 1]$ denotes flow-matching time, $\tau \in [0, \mathcal{T}]$ denotes physical PDE time, and $\lambda \in [0, 1]$ denotes the normalized coordinate used by the endpoint Hermite surrogate. The current flow state is stored in standardized model coordinates; an endpoint is first predicted in those coordinates and then inverse-standardized before observations and physical residuals are evaluated.

The section proceeds from the learned prior to the implemented sampler. Section 2.1 defines the conditional path, marginal velocity, and endpoint predictor. Section 2.2 introduces the observation and physical guidance loss using the same component notation as the experimental specification. Section 2.3 formalizes the endpoint-only temporal surrogate, Section 2.4 derives the exact endpoint tilt and its plug-in approximation, and Section 2.5 gives the deterministic, stochastic, and hybrid finite-step updates analyzed later.

2.1 Flow matching and endpoint prediction

Following the conditional flow-matching construction of Lipman et al. (2023, 2024), let d be the number of scalar entries in a vectorized standardized PDE pair, let I_d be the $d \times d$ identity matrix, and draw independent random variables $X_0 \sim \pi_0 = \mathcal{N}(0, I_d)$ and $X_1 \sim \pi_1$. Here π_1 is the endpoint probability law of standardized PDE data pairs in $\mathbb{R}^d$; it need not possess a Lebesgue density. We consider the affine conditional path

$$X_t = \alpha_t X_1 + \sigma_t X_0, \quad \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1) = \mathcal{N}(\alpha_t \mathbf{x}_1, \sigma_t^2 I_d), \quad (1)$$

where α_t and σ_t are differentiable scalar schedules with $\alpha_0 = 0$, $\sigma_0 = 1$, $\alpha_1 = 1$, and $\sigma_1 = 0$. We assume $\alpha_t > 0$ and $\sigma_t > 0$ for $0 < t < 1$, with σ_t interpreted as the conditional standard

deviation. A dot over α_t or σ_t denotes differentiation with respect to flow time t . For $0 < t < 1$, Gaussian smoothing gives X_t a marginal density, denoted by π_t . Its conditional score is

$$\nabla_{\mathbf{x}} \log \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1) = -\frac{\mathbf{x} - \alpha_t \mathbf{x}_1}{\sigma_t^2}.$$

At fixed (X_0, X_1) , differentiation of Equation (1) gives the conditional velocity. We write this velocity as v_t (Lipman et al., 2024):

$$v_t(\mathbf{x} \mid \mathbf{x}_1) = a_t^{\text{fm}} \mathbf{x} + b_t^{\text{fm}} \nabla_{\mathbf{x}} \log \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1), \quad (2)$$

with

$$a_t^{\text{fm}} = \frac{\dot{\alpha}_t}{\alpha_t}, \quad b_t^{\text{fm}} = \frac{\dot{\alpha}_t \sigma_t^2 - \dot{\sigma}_t \sigma_t \alpha_t}{\alpha_t}.$$

The marginal velocity is the posterior average of the conditional velocity:

$$v_t(\mathbf{x}) = \mathbb{E}[v_t(\mathbf{x} \mid X_1) \mid X_t = \mathbf{x}].$$

For the Conditional Optimal Transport (CondOT) path $\alpha_t = t$ and $\sigma_t = 1 - t$,

$$a_t^{\text{fm}} = \frac{1}{t}, \quad b_t^{\text{fm}} = \frac{1-t}{t}, \quad \nabla_{\mathbf{x}} \log \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1) = -\frac{\mathbf{x} - t\mathbf{x}_1}{(1-t)^2}, \quad v_t(\mathbf{x} \mid \mathbf{x}_1) = \frac{\mathbf{x}_1 - \mathbf{x}}{1-t}.$$

The network $v_t^\theta(\mathbf{x}) \equiv v_\theta(\mathbf{x}, t)$, parameterized by trainable weights θ , is trained by conditional flow matching,

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E} \left[\|v_t^\theta(X_t) - v_t(X_t \mid X_1)\|_2^2 \right],$$

where the expectation is over X_0 , X_1 , and the flow-time sampling distribution reported in Section C. For the PDE data in this study, to ensure training stability, we performed channel-wise normalization on the data in the original physical space using the mean and standard deviation of the training set. All training was conducted in the normalized model space.

For the CondOT path, the Euler endpoint predictor in standardized model coordinates is

$$P_t^\theta(\mathbf{x}) = \mathbf{x} + (1-t)v_t^\theta(\mathbf{x}), \quad P_1^\theta(\mathbf{x}) = \mathbf{x}. \quad (3)$$

For a midpoint step, P_t^θ denotes the corresponding differentiable predictor formed from the midpoint state and velocity. Let $\mathcal{D}_{\text{phys}}$ denote the checkpoint-defined channel-wise inverse standardization followed by the fixed channel split into physical fields. The decoded endpoint map is

$$E_t^\theta(\mathbf{x}) = \mathcal{D}_{\text{phys}}(P_t^\theta(\mathbf{x})).$$

The distinction between $P_t^\theta(\mathbf{x})$ and $E_t^\theta(\mathbf{x})$ lies in the fact that the former originates from the output of flow matching and resides in the normalized model space, whereas the latter corresponds to a quantity in the actual physical space. For a differentiable map f , let J_f denote its Jacobian and let $\|J_f\|_{\text{op}}$ denote the corresponding Euclidean operator norm. All observation and physical losses are evaluated on $E_t^\theta(\mathbf{x})$ in physical units.

2.2 Observation and physical guidance objective

We first introduce the common notation used by the method and the analysis. Let $\mathcal{Z}$ denote the finite-dimensional physical endpoint space. To distinguish between model space and physical space, we retain $\mathbf{x}$ to denote the result in the normalized model space and use $\mathbf{z} = (\mathbf{a}, \mathbf{u}) \in \mathcal{Z}$ to denote the corresponding result in physical space. Depending on the PDE, $\mathbf{a}$ is a coefficient, source term, or initial state, whereas $\mathbf{u}$ is a steady solution, terminal state, or full space-time field. Let $\mathcal{M}_{\mathbf{a}}$ and $\mathcal{M}_{\mathbf{u}}$ be the binary observation masks after broadcasting to the corresponding physical channels, and let $\mathcal{O} = (\mathcal{O}_{\mathbf{a}}, \mathcal{O}_{\mathbf{u}})$ be the observed values. With $m_{\mathbf{a}} = \|\mathcal{M}_{\mathbf{a}}\|_1$ and $m_{\mathbf{u}} = \|\mathcal{M}_{\mathbf{u}}\|_1$, nonempty observation components use

$$\begin{aligned}\mathcal{L}_{\text{obs},\mathbf{a}}(\mathbf{z}) &= \frac{1}{m_{\mathbf{a}}} \|\mathcal{M}_{\mathbf{a}} \odot (\mathbf{a} - \mathcal{O}_{\mathbf{a}})\|_2^2, \\ \mathcal{L}_{\text{obs},\mathbf{u}}(\mathbf{z}) &= \frac{1}{m_{\mathbf{u}}} \|\mathcal{M}_{\mathbf{u}} \odot (\mathbf{u} - \mathcal{O}_{\mathbf{u}})\|_2^2.\end{aligned}\tag{4}$$

Here $\odot$ denotes elementwise multiplication and, because the masks are binary, $\|\mathcal{M}_{\mathbf{a}}\|_1$ and $\|\mathcal{M}_{\mathbf{u}}\|_1$ counts active scalar entries after channel broadcasting. The physical loss follows:

$$\mathcal{L}_{\text{PDE},h}(\mathbf{z}) = \mathcal{L}_{\text{int},h}(\mathbf{z}) + \lambda_{\text{bc}}\mathcal{L}_{\text{bc},h}(\mathbf{z}) + \lambda_{\text{ic}}\mathcal{L}_{\text{ic},h}(\mathbf{z}) + \lambda_{\text{ep}}\mathcal{L}_{\text{ep},h}(\mathbf{z}).\tag{5}$$

Each active term in Equation (5) is the mean squared value of its own residual vector and therefore has its own denominator. The label int denotes the interior governing-equation residual, bc the boundary-condition residual, ic the initial-condition residual, and ep the endpoint-consistency residual used by endpoint-only temporal formulations. h denotes the spatial discretization parameter or, more generally, the collection of grid spacings and discrete spatial operators entering the residual.

Based on Equations (4) and (5), the complete endpoint guidance loss is

$$\mathcal{L}_h(\mathbf{z}) = \zeta_{\mathbf{a}}\mathcal{L}_{\text{obs},\mathbf{a}}(\mathbf{z}) + \zeta_{\mathbf{u}}\mathcal{L}_{\text{obs},\mathbf{u}}(\mathbf{z}) + \zeta_{\text{pde}}\mathcal{L}_{\text{PDE},h}(\mathbf{z}),\tag{6}$$

where $\zeta_{\mathbf{a}}$, $\zeta_{\mathbf{u}}$, and ζ_{pde} are the input-side observation, output-side observation, and physical-residual weights. Define the active component-label set

$$\mathcal{I} \subseteq \{\text{obs}_{\mathbf{a}}, \text{obs}_{\mathbf{u}}, \text{int}, \text{bc}, \text{ic}, \text{ep}\}.$$

The set $\mathcal{I}$ contains only components enabled by the task, masks, and residual configuration. For $j \in \mathcal{I}$, let $R_{h,j} : \mathcal{Z} \rightarrow \mathbb{R}^{m_j}$ be the corresponding residual vector and let m_j be its number of active scalar entries. For observation labels, the subscript h is only part of the unified notation; those residuals do not use a spatial differential operator. Then Equation (6) is equivalently

$$\mathcal{L}_h(\mathbf{z}) = \sum_{j \in \mathcal{I}} \frac{w_j}{m_j} \|R_{h,j}(\mathbf{z})\|_2^2,\tag{7}$$

with

$$\begin{aligned}w_{\text{obs}_{\mathbf{a}}} &= \zeta_{\mathbf{a}}, & w_{\text{obs}_{\mathbf{u}}} &= \zeta_{\mathbf{u}}, & w_{\text{int}} &= \zeta_{\text{pde}}, \\ w_{\text{bc}} &= \zeta_{\text{pde}}\lambda_{\text{bc}}, & w_{\text{ic}} &= \zeta_{\text{pde}}\lambda_{\text{ic}}, & w_{\text{ep}} &= \zeta_{\text{pde}}\lambda_{\text{ep}}.\end{aligned}$$

A zero-weight or empty component is omitted from $\mathcal{I}$.

Generalized endpoint-posterior interpretation. Equation (7) can be interpreted as a generalized Bayesian loss update in the sense of Bissiri et al. (2016). The component label j is not a spatial or channel index. Define

$$\mathcal{I}_{\text{obs}} = \mathcal{I} \cap \{\text{obs}_{\mathbf{a}}, \text{obs}_{\mathbf{u}}\}, \quad \mathcal{I}_{\text{phys}} = \mathcal{I} \cap \{\text{int}, \text{bc}, \text{ic}, \text{ep}\}.$$

For $j \in \mathcal{I}_{\text{obs}}$, let $Y_j \in \mathbb{R}^{m_j}$ collect the observed scalar entries and let $\mathcal{O}_j : \mathcal{Z} \rightarrow \mathbb{R}^{m_j}$ be the corresponding observation operator. A Gaussian observation model is

$$Y_j = \mathcal{O}_j(\mathbf{z}) + \epsilon_j^{\text{obs}}, \quad \epsilon_j^{\text{obs}} \sim \mathcal{N}(0, \sigma_j^2 I_{m_j}), \quad j \in \mathcal{I}_{\text{obs}},$$

where I_{m_j} is the $m_j \times m_j$ identity matrix, with residual $R_{h,j}(\mathbf{z}) = \mathcal{O}_j(\mathbf{z}) - Y_j$. For a physical component $j \in \mathcal{I}_{\text{phys}}$, the formal pseudo-observation model is

$$\mathbf{0}_{m_j} = R_{h,j}(\mathbf{z}) + \epsilon_j^{\text{phys}}, \quad \epsilon_j^{\text{phys}} \sim \mathcal{N}(0, \sigma_j^2 I_{m_j}), \quad j \in \mathcal{I}_{\text{phys}}.$$

Here $\mathbf{0}_{m_j}$ is the zero vector in $\mathbb{R}^{m_j}$. The latter is an energy-based interpretation of a discrete physical constraint, not a claim that the PDE residual is independently measured. If the error vectors are independent across active components and scalar entries, then, for fixed m_j and $w_j > 0$, the choice $\sigma_j^2 = m_j/2w_j$ makes the negative log-likelihood of component j equal to $w_j \|R_{h,j}(\mathbf{z})\|_2^2/m_j$, up to a constant independent of $\mathbf{z}$. Consequently, Equation (7) defines an endpoint energy on decoded physical fields.

2.3 Endpoint-only temporal residuals

For endpoint-pair datasets, the generated sample contains an initial state and a terminal state but not an arbitrary intermediate trajectory. The default temporal residual therefore uses a cubic Hermite curve induced by the two endpoints and the semidiscrete PDE right-hand side. This construction is an endpoint surrogate.

Semidiscrete physical state. Fix the spatial discretization parameter h . Let $\mathbf{s}_h(\tau) \in \mathbb{R}^{d_h}$ denote the spatially discretized physical state at physical time $\tau \in [0, \mathcal{T}]$, where d_h is the number of scalar spatial degrees of freedom across all state channels. For fixed PDE coefficients and physical parameters, suppressed from the notation, the semidiscrete dynamics are

$$\dot{\mathbf{s}}_h(\tau) = \mathcal{G}_h(\mathbf{s}_h(\tau)), \quad 0 \leq \tau \leq \mathcal{T}. \quad (8)$$

Thus, $\mathbf{s}_h$ is the discretized physical state and $\dot{\mathbf{s}}_h$ is its derivative with respect to physical time. The map $\mathcal{G}_h : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_h}$ is the spatially discretized PDE right-hand side.

Write $\mathbf{s}_{h,0} = \mathbf{s}_h(0)$ and $\mathbf{s}_{h,\mathcal{T}} = \mathbf{s}_h(\mathcal{T})$. For $\lambda \in [0, 1]$, define the endpoint Hermite curve

$$\mathcal{H}_h(\lambda) = h_{00}(\lambda)\mathbf{s}_{h,0} + h_{10}(\lambda)\mathcal{T}\mathcal{G}_h(\mathbf{s}_{h,0}) + h_{01}(\lambda)\mathbf{s}_{h,\mathcal{T}} + h_{11}(\lambda)\mathcal{T}\mathcal{G}_h(\mathbf{s}_{h,\mathcal{T}}),$$

where

$$\begin{aligned} h_{00}(\lambda) &= 2\lambda^3 - 3\lambda^2 + 1, & h_{10}(\lambda) &= \lambda^3 - 2\lambda^2 + \lambda, \\ h_{01}(\lambda) &= -2\lambda^3 + 3\lambda^2, & h_{11}(\lambda) &= \lambda^3 - \lambda^2. \end{aligned}$$

The same curve parameterized by physical time is

$$\tilde{\mathcal{H}}_h(\tau) := \mathcal{H}_h\left(\frac{\tau}{\mathcal{T}}\right), \quad 0 \leq \tau \leq \mathcal{T}.$$

The implemented interior residual evaluates $\dot{\tilde{\mathcal{H}}}_h - \mathcal{G}_h(\tilde{\mathcal{H}}_h)$ at prescribed physical-time collocation points. The optional endpoint component uses the trapezoidal defect in Equation (11) below. The proposition that follows is a fixed-grid application of standard cubic Hermite interpolation and trapezoidal quadrature error identities; see, for example, Hairer et al. (1993).

Proposition 1 (Fixed-grid consistency of the endpoint Hermite surrogate) *Fix h and $\mathcal{T} > 0$. Let $U_h \subseteq \mathbb{R}^{d_h}$ be open and assume*

$$\mathcal{G}_h \in C^3(U_h; \mathbb{R}^{d_h}).$$

Let $\mathbf{s}_h : [0, \mathcal{T}] \rightarrow U_h$ solve Equation (8). Assume that the compact tube joining the exact and Hermite paths is compactly contained in U_h , denoted by $K_h^{\text{tube}} \Subset U_h$:

$$K_h^{\text{tube}} := \left\{ (1 - \vartheta)\mathbf{s}_h(\tau) + \vartheta\tilde{\mathcal{H}}_h(\tau) : \tau \in [0, \mathcal{T}], \vartheta \in [0, 1] \right\} \Subset U_h.$$

Then $\mathbf{s}_h \in C^4([0, \mathcal{T}]; \mathbb{R}^{d_h})$. Writing $D\mathcal{G}_h$ for the Jacobian of $\mathcal{G}_h$, the finite constants

$$M_3 := \max_{0 \leq \tau \leq \mathcal{T}} \|\mathbf{s}_h^{(3)}(\tau)\|_2,$$

$$M_4 := \max_{0 \leq \tau \leq \mathcal{T}} \|\mathbf{s}_h^{(4)}(\tau)\|_2,$$

$$L_{\mathcal{G}} := \max_{\mathbf{z} \in K_h^{\text{tube}}} \|D\mathcal{G}_h(\mathbf{z})\|_{\text{op}}$$

satisfy

$$\sup_{0 \leq \tau \leq \mathcal{T}} \|\tilde{\mathcal{H}}_h(\tau) - \mathbf{s}_h(\tau)\|_2 \leq \frac{1}{4} M_4 \mathcal{T}^4, \quad (9)$$

$$\sup_{0 \leq \tau \leq \mathcal{T}} \left\| \dot{\tilde{\mathcal{H}}}_h(\tau) - \mathcal{G}_h(\tilde{\mathcal{H}}_h(\tau)) \right\|_2 \leq \frac{19}{48} M_4 \mathcal{T}^3 + \frac{1}{4} L_{\mathcal{G}} M_4 \mathcal{T}^4, \quad (10)$$

$$\left\| \mathbf{s}_{h,\mathcal{T}} - \mathbf{s}_{h,0} - \frac{\mathcal{T}}{2} \{ \mathcal{G}_h(\mathbf{s}_{h,0}) + \mathcal{G}_h(\mathbf{s}_{h,\mathcal{T}}) \} \right\|_2 \leq \frac{M_3 \mathcal{T}^3}{12}. \quad (11)$$

Consequently, when the local constants remain bounded over a family of short time horizons, the Hermite state error is $O(\mathcal{T}^4)$ and the PDE and endpoint-integral defects are $O(\mathcal{T}^3)$. The statement is relative to the fixed semidiscrete operator $\mathcal{G}_h$; it is neither a uniform-in- h result nor a claim that two endpoints uniquely determine a nonlinear trajectory.

Proof Because $\mathcal{G}_h \in C^3(U_h)$ and $\dot{\mathbf{s}}_h = \mathcal{G}_h(\mathbf{s}_h)$, repeated differentiation of the ODE implies $\mathbf{s}_h \in C^4$. Continuity on the compact interval $[0, \mathcal{T}]$ gives $M_3, M_4 < \infty$. The set K_h^{tube} is compact, and $D\mathcal{G}_h$ is continuous on U_h , hence $L_{\mathcal{G}} < \infty$.

Let P_3 be the third-order Taylor polynomial of $\mathbf{s}_h$ at $\tau = 0$ and write

$$\mathbf{s}_h(\tau) = P_3(\tau) + r(\tau).$$

The vector-valued integral remainder is

$$\begin{aligned} r(\tau) &= \int_0^\tau \frac{(\tau-s)^3}{6} \mathbf{s}_h^{(4)}(s) \, ds, \\ \dot{r}(\tau) &= \int_0^\tau \frac{(\tau-s)^2}{2} \mathbf{s}_h^{(4)}(s) \, ds. \end{aligned}$$

Therefore

$$\|r(\tau)\|_2 \leq \frac{M_4 \tau^4}{24}, \quad \|\dot{r}(\tau)\|_2 \leq \frac{M_4 \tau^3}{6}. \quad (12)$$

Cubic Hermite interpolation reproduces P_3 exactly. Since $r(0) = \dot{r}(0) = 0$, the Hermite interpolant of the remainder is

$$\tilde{\mathcal{H}}_{h,r}(\tau) = h_{01} \left(\frac{\tau}{\mathcal{T}} \right) r(\mathcal{T}) + h_{11} \left(\frac{\tau}{\mathcal{T}} \right) \mathcal{T} \dot{r}(\mathcal{T}).$$

On $[0, 1]$,

$$|h_{01}| \leq 1, \quad |h_{11}| \leq 1, \quad |h'_{01}| \leq \frac{3}{2}, \quad |h'_{11}| \leq 1. \quad (13)$$

Equations (12)- (13) yield

$$\begin{aligned} \|\tilde{\mathcal{H}}_{h,r}(\tau)\|_2 &\leq \frac{5}{24} M_4 \mathcal{T}^4, \\ \|\dot{\tilde{\mathcal{H}}}_{h,r}(\tau)\|_2 &\leq \frac{11}{48} M_4 \mathcal{T}^3. \end{aligned}$$

Since $\tilde{\mathcal{H}}_h - \mathbf{s}_h = \tilde{\mathcal{H}}_{h,r} - r$, the triangle inequality gives

$$\|\tilde{\mathcal{H}}_h(\tau) - \mathbf{s}_h(\tau)\|_2 \leq \left(\frac{5}{24} + \frac{1}{24} \right) M_4 \mathcal{T}^4 = \frac{1}{4} M_4 \mathcal{T}^4, \quad (14)$$

$$\|\dot{\tilde{\mathcal{H}}}_h(\tau) - \dot{\mathbf{s}}_h(\tau)\|_2 \leq \left(\frac{11}{48} + \frac{1}{6} \right) M_4 \mathcal{T}^3 = \frac{19}{48} M_4 \mathcal{T}^3. \quad (15)$$

This proves Equation (9).

For each τ , the segment between $\mathbf{s}_h(\tau)$ and $\tilde{\mathcal{H}}_h(\tau)$ belongs to K_h^{tube} . The fundamental theorem of calculus along this segment therefore gives

$$\begin{aligned} &\mathcal{G}_h(\tilde{\mathcal{H}}_h(\tau)) - \mathcal{G}_h(\mathbf{s}_h(\tau)) \\ &= \int_0^1 D\mathcal{G}_h \left(\mathbf{s}_h(\tau) + \vartheta \{ \tilde{\mathcal{H}}_h(\tau) - \mathbf{s}_h(\tau) \} \right) \{ \tilde{\mathcal{H}}_h(\tau) - \mathbf{s}_h(\tau) \} \, d\vartheta. \end{aligned}$$

Hence

$$\|\mathcal{G}_h(\tilde{\mathcal{H}}_h(\tau)) - \mathcal{G}_h(\mathbf{s}_h(\tau))\|_2 \leq L_{\mathcal{G}} \|\tilde{\mathcal{H}}_h(\tau) - \mathbf{s}_h(\tau)\|_2. \quad (16)$$

Using $\dot{\mathbf{s}}_h = \mathcal{G}_h(\mathbf{s}_h)$ and adding and subtracting $\dot{\mathbf{s}}_h$ gives

$$\dot{\tilde{\mathcal{H}}}_h - \mathcal{G}_h(\tilde{\mathcal{H}}_h) = \{ \dot{\tilde{\mathcal{H}}}_h - \dot{\mathbf{s}}_h \} + \{ \mathcal{G}_h(\mathbf{s}_h) - \mathcal{G}_h(\tilde{\mathcal{H}}_h) \}.$$

Equations (14), (15), and (16) prove Equation (10).

Finally,

$$\mathbf{s}_{h,\mathcal{T}} - \mathbf{s}_{h,0} = \int_0^{\mathcal{T}} \dot{\mathbf{s}}_h(\tau) \, d\tau. \quad (17)$$

For any $f \in C^2([0, \mathcal{T}]; \mathbb{R}^{d_h})$, integration by parts gives the vector-valued trapezoidal identity

$$\int_0^{\mathcal{T}} f(\tau) \, d\tau - \frac{\mathcal{T}}{2} \{f(0) + f(\mathcal{T})\} = -\frac{1}{2} \int_0^{\mathcal{T}} \tau(\mathcal{T} - \tau) f''(\tau) \, d\tau. \quad (18)$$

Apply Equation (18) with $f = \dot{\mathbf{s}}_h$. Since $f'' = \mathbf{s}_h^{(3)}$ and $\int_0^{\mathcal{T}} \tau(\mathcal{T} - \tau) \, d\tau = \mathcal{T}^3/6$, Equation (11) follows. $\blacksquare$

Remark 2 (Applicability to the temporal PDE families) *The proposition uses local, fixed-grid regularity. For Heat, the first-order Wave system, and Advection-Diffusion with fixed coefficients, the semidiscrete right-hand side is linear and therefore smooth globally at fixed h . The Reaction-Diffusion right-hand side is polynomial and smooth but only locally Lipschitz on bounded states. In the corrected periodic Navier-Stokes formulation, FFT differentiation, the zero-mean projection, and the stream-function solve are linear at fixed resolution, while the advection term is quadratic in vorticity; the resulting finite-dimensional map is smooth on bounded vorticity sets. The constants may nevertheless grow with resolution, high-frequency content, decreasing viscosity, or time horizon. For Shallow Water, the smoothness assumption applies only in a wet-state region separated from zero depth and from the switching surface of any safe-depth clamp or nonsmooth flux safeguard. Burgers uses a stored full space-time field in the main implementation, so its principal residual does not use this endpoint surrogate.*

Remark 3 (Operator-relative consistency) *The bounds compare the Hermite surrogate with a trajectory generated by the same semidiscrete operator $\mathcal{G}_h$ used in the residual. If the data are generated by a different numerical solver, the discrepancy between the two discrete dynamics is an additional model-discretization term. This term is included later in the physical consistency error but not in Equations (9)-(11).*

2.4 Exact endpoint tilt and plug-in guidance

The guidance loss in Equation (6) can be used to form an energy tilt of the learned endpoint law. Define the tilted probability measure π_1^{obs} by its Radon-Nikodym derivative with respect to π_1 :

$$\frac{d\pi_1^{\text{obs}}}{d\pi_1}(\mathbf{x}_1) = \frac{1}{\mathfrak{Z}_{\text{obs}}} \exp\{-\mathcal{L}_h(\mathcal{D}_{\text{phys}}(\mathbf{x}_1))\},$$

$$\mathfrak{Z}_{\text{obs}} = \int e^{-\mathcal{L}_h(\mathcal{D}_{\text{phys}}(\mathbf{x}_1))} \pi_1(d\mathbf{x}_1).$$

We assume $0 < \mathfrak{Z}_{\text{obs}} < \infty$. The superscript obs records the dependence on the observations and physical constraints entering $\mathcal{L}_h$; it does not imply that every component is a literal observation likelihood. The next result identifies the exact change in the intermediate-time marginal and thereby separates the probabilistic conditioning operation from the point-estimate approximation used by the sampler.

Proposition 4 (Exact tilted marginal and velocity identities) *Fix $t \in (0, 1)$ and let the original joint law of (X_t, X_1) be induced by Equation (1) and the endpoint law π_1 . Define*

$$W(X_1) = \exp\{-\mathcal{L}_h(\mathcal{D}_{\text{phys}}(X_1))\}, \quad \Lambda_t(\mathbf{x}) = \mathbb{E}[W(X_1) \mid X_t = \mathbf{x}]. \quad (19)$$

Assume that $\Lambda_t(\mathbf{x}) > 0$, that the relevant densities are differentiable, and that differentiation can be interchanged with integration over $\mathbf{x}_1$. Then the marginal density π_t^{obs} generated by the tilted endpoint law π_1^{obs} satisfies

$$\pi_t^{\text{obs}}(\mathbf{x}) = \frac{\pi_t(\mathbf{x})\Lambda_t(\mathbf{x})}{\mathfrak{Z}_{\text{obs}}}, \quad (20)$$

$$\nabla_{\mathbf{x}} \log \pi_t^{\text{obs}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \pi_t(\mathbf{x}) + \nabla_{\mathbf{x}} \log \Lambda_t(\mathbf{x}). \quad (21)$$

If v_t and v_t^{obs} are the marginal velocity fields associated with the original and tilted endpoint laws through the same conditional path, then

$$v_t^{\text{obs}}(\mathbf{x}) - v_t(\mathbf{x}) = b_t^{\text{fm}} \nabla_{\mathbf{x}} \log \Lambda_t(\mathbf{x}). \quad (22)$$

Proof Marginalizing the conditional path against the tilted endpoint law gives

$$\pi_t^{\text{obs}}(\mathbf{x}) = \int \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1) \pi_1^{\text{obs}}(d\mathbf{x}_1) \quad (23)$$

$$= \frac{1}{\mathfrak{Z}_{\text{obs}}} \int \pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1) W(\mathbf{x}_1) \pi_1(d\mathbf{x}_1). \quad (24)$$

A regular conditional endpoint law under the original joint distribution is

$$\pi_1(d\mathbf{x}_1 \mid X_t = \mathbf{x}) = \frac{\pi_{t|1}(\mathbf{x} \mid \mathbf{x}_1)}{\pi_t(\mathbf{x})} \pi_1(d\mathbf{x}_1).$$

Substitution into Equation (24) gives

$$\pi_t^{\text{obs}}(\mathbf{x}) = \frac{\pi_t(\mathbf{x})}{\mathfrak{Z}_{\text{obs}}} \int W(\mathbf{x}_1) \pi_1(d\mathbf{x}_1 \mid X_t = \mathbf{x}) = \frac{\pi_t(\mathbf{x})\Lambda_t(\mathbf{x})}{\mathfrak{Z}_{\text{obs}}},$$

which proves Equation (20). Taking logarithms and differentiating proves Equation (21).

Under the stated differentiation-integration interchange, the conditional-score identity

$$\mathbb{E}[\nabla_{\mathbf{x}} \log \pi_{t|1}(\mathbf{x} \mid X_1) \mid X_t = \mathbf{x}] = \nabla_{\mathbf{x}} \log \pi_t(\mathbf{x})$$

holds under the original endpoint law. Averaging Equation (2) under the corresponding endpoint posterior therefore gives

$$v_t(\mathbf{x}) = a_t^{\text{fm}} \mathbf{x} + b_t^{\text{fm}} \nabla_{\mathbf{x}} \log \pi_t(\mathbf{x}).$$

The same conditional path combined with π_1^{obs} gives

$$v_t^{\text{obs}}(\mathbf{x}) = a_t^{\text{fm}} \mathbf{x} + b_t^{\text{fm}} \nabla_{\mathbf{x}} \log \pi_t^{\text{obs}}(\mathbf{x}).$$

Subtracting these identities and applying Equation (21) proves Equation (22). ■

For the CondOT path, the exact marginal velocity obeys the conditional-mean identity

$$\mathbf{x} + (1 - t)v_t(\mathbf{x}) = \mathbb{E}[X_1 \mid X_t = \mathbf{x}].$$

This identity alone does not reduce the conditional expectation in Equation (19) to an energy evaluated at a single endpoint: in general,

$$\log \mathbb{E} \left[e^{-\mathcal{L}_h(\mathcal{D}_{\text{phys}}(X_1))} \mid X_t = \mathbf{x} \right] \neq -\mathcal{L}_h(\mathcal{D}_{\text{phys}}(\mathbb{E}[X_1 \mid X_t = \mathbf{x}])).$$

FM4PDE replaces the conditional endpoint law by the differentiable endpoint prediction $P_t^\theta(\mathbf{x})$ and uses the decoded physical endpoint $E_t^\theta(\mathbf{x}) = \mathcal{D}_{\text{phys}}(P_t^\theta(\mathbf{x}))$. This gives the plug-in approximation

$$\nabla_{\mathbf{x}} \log \Lambda_t(\mathbf{x}) \approx -\nabla_{\mathbf{x}} \mathcal{L}_h(E_t^\theta(\mathbf{x})) = -J_{E_t^\theta}(\mathbf{x})^\top \nabla_{\mathbf{z}} \mathcal{L}_h(E_t^\theta(\mathbf{x})). \quad (25)$$

Thus, the observation operators and discrete PDE operators act on a denoised, decoded endpoint in physical units. The chain rule then transports their gradient through both the affine inverse-standardization map and the learned endpoint predictor to the current flow state.

For later reference, define the plug-in and velocity approximation errors by

$$\begin{aligned} \varepsilon_t^{\text{plug}}(\mathbf{x}) &= \nabla_{\mathbf{x}} \log \Lambda_t(\mathbf{x}) + \nabla_{\mathbf{x}} \mathcal{L}_h(E_t^\theta(\mathbf{x})), \\ \varepsilon_t^{\text{vel}}(\mathbf{x}) &= v_t^\theta(\mathbf{x}) - v_t(\mathbf{x}). \end{aligned}$$

The implemented deterministic drift and the exact tilted drift are related by

$$\left[v_t^\theta(\mathbf{x}) - b_t^{\text{fm}} \nabla_{\mathbf{x}} \mathcal{L}_h(E_t^\theta(\mathbf{x})) \right] - v_t^{\text{obs}}(\mathbf{x}) = \varepsilon_t^{\text{vel}}(\mathbf{x}) - b_t^{\text{fm}} \varepsilon_t^{\text{plug}}(\mathbf{x}).$$

Accordingly, the endpoint tilt supplies a probabilistic target, whereas the implemented sampler is an inexact finite-step realization of its drift. A constant common multiplier can be absorbed into $\mathcal{L}_h$ and therefore corresponds to another fixed endpoint tilt. A flow-time-dependent multiplier is instead an algorithmic tempering schedule and need not generate the marginals of any single fixed generalized posterior. The analysis below is therefore formulated directly for the implemented update, with proposal and endpoint-transport discrepancies retained as explicit defect terms.

2.5 Implemented guided proposals

Let $0 = t_0 < t_1 < \dots < t_N = 1$ and $\Delta t_k = t_{k+1} - t_k$. Put $F_t(\mathbf{x}) = \mathcal{L}_h(E_t^\theta(\mathbf{x}))$, $g_t(\mathbf{x}) = \nabla_{\mathbf{x}} F_t(\mathbf{x})$. The reference configuration uses endpoint loss and the current-state chain rule, so g_t contains the Jacobian in Equation (25). Both deterministic and stochastic phases use this same endpoint objective. The Euler deterministic proposal and the stochastic re-initialization proposal are

$$\begin{aligned} \mathcal{A}_k^{\text{D}}(\mathbf{x}) &= \mathbf{x} + \Delta t_k v_{t_k}^\theta(\mathbf{x}), \\ \mathcal{A}_k^{\text{S}}(\mathbf{x}, \xi_k) &= (1 - t_{k+1})\xi_k + t_{k+1} P_{t_k}^\theta(\mathbf{x}), \quad \xi_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d). \end{aligned}$$

The midpoint option changes only the construction of the differentiable proposal and endpoint predictor.

The main theoretical statements cover global-norm clipping, which is applied after the weighted component gradients have been summed:

$$\mathcal{C}_{G_c}(v) = \rho(v)v, \quad \rho(v) = \min \left\{ 1, \frac{G_c}{\|v\|_2 + \varepsilon_{\text{clip}}} \right\}, \quad \varepsilon_{\text{clip}} > 0. \quad (26)$$

Per-component clipping is an ablation that generally changes the direction of the total gradient and is not covered by the scalar-alignment results below.

Let γ_k be the update multiplier returned by the configured sampler. In the analyzed default branches,

$$\gamma_k^{\text{D}} = \Delta t_k b_{t_k}^{\text{fm}}, \quad \gamma_k^{\text{S}} = c_\zeta(1 - t_k).$$

Both deterministic and stochastic steps can then be written as

$$\begin{aligned} \mathbf{y}_{k+1} &= \mathcal{A}_k(\mathbf{x}_k, \xi_k), \\ \mathbf{x}_{k+1} &= \mathbf{y}_{k+1} - \gamma_k \mathcal{C}_{G_c}(g_k) = \mathbf{y}_{k+1} - a_k g_k, \\ a_k &= \gamma_k \rho_k, \quad \rho_k = \rho(g_k), \quad g_k = \nabla F_{t_k}(\mathbf{x}_k). \end{aligned} \quad (27)$$

A hybrid sampler chooses $\mathcal{A}_k^{\text{D}}$ or $\mathcal{A}_k^{\text{S}}$ according to a prescribed switch index.

At $t_0 = 0$, the implementation evaluates $b_t^{\text{fm}} = (1-t)/t$ after replacing t by $\max\{t, 10^{-6}\}$. This clamp is a numerical safeguard, not a proof of stability. The deterministic theorems include the first correction only when its realized effective step satisfies the stated descent condition; otherwise the first step is treated as a finite initialization and the rate statement starts at a positive flow time. The default stochastic branch does not use the singular deterministic multiplier.

3 Conditional finite-step theory

This section analyzes the finite-step sampler in Equation (27). The argument is organized in four stages. First, the bounded clipped correction and a finite-horizon growth condition on the raw proposal yield a deterministic high-probability localization region. Second, the component-wise MSE structure and the endpoint map provide local smoothness constants on that region. Third, the change of endpoint objective across a raw proposal is represented by two observable transport defects. These ingredients give a one-step inexact descent inequality without a PL condition. Finally, a local PL inequality, when available, converts this descent inequality into contraction toward an explicit defect floor. The deterministic, stochastic, and hybrid results are all consequences of this common recursion.

The remainder of this section develops a finite-step analysis of the implemented sampler in Equation (27). The analysis is local, finite-dimensional, and finite-horizon: all variables are the finite-dimensional model coordinates induced by a fixed spatial grid, and all bounds are stated for a fixed sampling budget. Section 3.1 first constructs the high-probability localization region $\mathcal{K}_\delta$ from the finite-horizon proposal recursion and the bounded clipped correction. This region is an analysis domain obtained from the sampler dynamics; it is not an a priori assertion that every realization of the stochastic sampler is uniformly bounded. Section 3.2 then proves local smoothness of the implemented component-wise

Algorithm 1 Endpoint-guided FM4PDE sampling

Input: Pretrained velocity v_t^θ , decoder $\mathcal{D}_{\text{phys}}$, residual maps $\{R_{h,j}\}_{j \in \mathcal{I}}$, weights $\{w_j\}_{j \in \mathcal{I}}$, counts $\{m_j\}_{j \in \mathcal{I}}$, grid $0 = t_0 < t_1 < \dots < t_N = 1$, schedule s_k , stochastic coefficient c_ζ , clipping threshold G_c , and constants $\varepsilon_t, \varepsilon_{\text{clip}} > 0$.

- 1: $\mathbf{x}_0 \sim \mathcal{N}(0, I_d)$. ▷ Initialize in standardized model coordinates.
- 2: **for** $k = 0, \dots, N - 1$ **do**
- 3: $\Delta t_k = t_{k+1} - t_k$, $t_k^\varepsilon = \max\{t_k, \varepsilon_t\}$. ▷ Set the flow-time increment
- 4: $P_{t_k}^\theta(\mathbf{x}_k) = \mathbf{x}_k + (1 - t_k)v_{t_k}^\theta(\mathbf{x}_k)$. ▷ Predict the terminal endpoint
- 5: $\mathbf{z}_k = E_{t_k}^\theta(\mathbf{x}_k) = \mathcal{D}_{\text{phys}}(P_{t_k}^\theta(\mathbf{x}_k))$. ▷ Decode the endpoint prediction to physical unit
- 6: $\mathcal{L}_h(\mathbf{z}_k) = \sum_{j \in \mathcal{I}} w_j m_j^{-1} \|R_{h,j}(\mathbf{z}_k)\|_2^2$. ▷ Evaluate the guided losses with Equation (7)
- 7: $\mathbf{g}_k = \nabla_{\mathbf{x}_k} \mathcal{L}_h(E_{t_k}^\theta(\mathbf{x}_k))$ ▷ Backpropagate the endpoint loss
- 8: **if** Deterministic **then**
- 9: $\mathbf{y}_{k+1} = \mathcal{A}^D(\mathbf{x}_k) = \mathbf{x}_k + \Delta t_k v_{t_k}^\theta(\mathbf{x}_k)$. ▷ Deterministic proposal
- 10: $\gamma_k = \Delta t_k b_{t_k}^{\text{fm}}$ with $b_{t_k}^{\text{fm}} = (1 - t_k^\varepsilon)/t_k^\varepsilon$. ▷ CondOT deterministic guidance multiplier
- 11: **else**
- 12: $\xi_k \sim \mathcal{N}(0, I_d)$. ▷ Draw fresh stochastic reinjection noise
- 13: $\mathbf{y}_{k+1} = \mathcal{A}^S(\mathbf{x}_k, \xi_k) = (1 - t_{k+1})\xi_k + t_{k+1}P_{t_k}^\theta(\mathbf{x}_k)$. ▷ Stochastic proposal
- 14: $\gamma_k = c_\zeta(1 - t_k)$. ▷ Stochastic endpoint-guidance multiplier
- 15: **end if**
- 16: $C_{G_c}(\mathbf{g}_k) = \rho_k \mathbf{g}_k$ with $\rho_k = \min\{1, G_c/(\|\mathbf{g}_k\|_2 + \varepsilon_{\text{clip}})\}$. ▷ Clipping
- 17: $\mathbf{x}_{k+1} = \mathbf{y}_{k+1} - \gamma_k C_{G_c}(\mathbf{g}_k)$. ▷ Perform the guided update
- 18: **end for**
- 19: $\mathbf{z}_N = E_1^\theta(\mathbf{x}_N) = \mathcal{D}_{\text{phys}}(\mathbf{x}_N)$. ▷ Decode the final terminal sample
- 20: **return** $\mathbf{z}_N$.

MSE objective on this region, and Section 3.3 records sufficient conditions under which the local PL geometry used later is justified, while also emphasizing possible non-identifiability. Section 3.4 keeps the clipping factor explicitly in the effective guidance step, and Section 3.5 quantifies the discrepancy between the gradient evaluated before the raw proposal and the objective seen after the proposal. These ingredients are combined in Section 3.6 to obtain the basic inexact descent recursion. Sections 3.7–3.10 specialize this common recursion to the deterministic, stochastic, implementation-specific stochastic, and hybrid samplers, yielding contraction only above explicit defect floors. Finally, Section 3.11 translates the discrete guidance loss bound into a statement about proximity to a physical solution set under separate consistency and identifiability assumptions.

3.1 Finite-horizon localization

We begin with the part of the argument that is independent of the objective geometry. For $G_c > 0$, the global clipping operator in Equation (26) has the elementary properties below.

Lemma 5 (Geometry of global norm clipping) *For every $v \in \mathbb{R}^d$,*

$$\mathcal{C}_{G_c}(v) = \rho(v)v, \quad \langle v, \mathcal{C}_{G_c}(v) \rangle = \rho(v)\|v\|_2^2, \quad \|\mathcal{C}_{G_c}(v)\|_2 \leq G_c,$$

where $\rho(v)$ is defined in Equation (26). Hence clipping preserves the direction of a nonzero vector and bounds the magnitude of the applied guidance correction.

Proof The first two identities follow directly from the scalar definition of $\rho(v)$. Moreover,

$$\|\mathcal{C}_{G_c}(v)\|_2 = \min \left\{ \|v\|_2, \frac{G_c \|v\|_2}{\|v\|_2 + \varepsilon_{\text{clip}}} \right\} \leq G_c.$$

■

Assumption 1 (Finite-horizon proposal growth) For the fixed grid $0 = t_0 < \dots < t_N = 1$, there are deterministic constants $A_k, B_k \geq 0$ such that

$$\|\mathcal{A}_k(\mathbf{x}, \xi)\|_2 \leq A_k \|\mathbf{x}\|_2 + B_k(1 + \|\xi\|_2) \quad (28)$$

for all $\mathbf{x}, \xi \in \mathbb{R}^d$; deterministic steps use $\xi = 0$. In addition, $0 \leq \gamma_k \leq \bar{\gamma} < \infty$.

The global bound in Equation (28) can be replaced by a local version on the deterministic balls used in the localization recursion. Namely, it is sufficient that, for each recursively defined radius r_k , there exist constants $A_k(r_k)$ and $B_k(r_k)$ such that the same bound holds for all $\mathbf{x} \in \overline{B}(0, r_k)$ and all ξ . These constants may depend on the radius, the trained network, the fixed grid, N , and δ , but they must be fixed before the current Gaussian noise at step k is observed. This prevents the localization argument from choosing growth constants retrospectively from the realized trajectory.

Assumption 1 is a finite-horizon linear-growth condition. It holds for the Euler proposal when the learned velocity has linear growth on the relevant interval, and for the stochastic proposal when the endpoint predictor has linear growth.

Theorem 6 (Finite-horizon pathwise and moment stability) Under Assumption 1, the implemented globally clipped update satisfies

$$\|\mathbf{x}_{k+1}\|_2 \leq A_k \|\mathbf{x}_k\|_2 + B_k(1 + \|\xi_k\|_2) + \gamma_k G_c. \quad (29)$$

Consequently,

$$\|\mathbf{x}_k\|_2 \leq \left(\prod_{i=0}^{k-1} A_i \right) \|\mathbf{x}_0\|_2 + \sum_{j=0}^{k-1} \left(\prod_{i=j+1}^{k-1} A_i \right) \{B_j(1 + \|\xi_j\|_2) + \gamma_j G_c\}, \quad (30)$$

where an empty product equals one. If $p \geq 1$, define $M_{p,0} = (\mathbb{E} \|\mathbf{x}_0\|_2^p)^{1/p} < \infty$ and $\nu_{p,k} = (\mathbb{E} \|\xi_k\|_2^p)^{1/p} < \infty$. Then

$$M_{p,k} := (\mathbb{E} \|\mathbf{x}_k\|_2^p)^{1/p} \leq \left(\prod_{i=0}^{k-1} A_i \right) M_{p,0} + \sum_{j=0}^{k-1} \left(\prod_{i=j+1}^{k-1} A_i \right) \{B_j(1 + \nu_{p,j}) + \gamma_j G_c\}. \quad (31)$$

Thus every finite-order moment remains finite over a finite number of steps.

Proof By Lemma 5, the norm of the applied correction in Equation (27) is at most $\gamma_k G_c$. The triangle inequality and Equation (28) therefore give Equation (29). Iterating this scalar inequality gives Equation (30). Taking L^p norms, applying Minkowski's inequality at each step, and iterating the resulting deterministic recursion proves Equation (31). $\blacksquare$

We next turn the pathwise estimate into a deterministic high-probability analysis region. Let $\Xi_1, \dots, \Xi_{N_G}$ enumerate all independent standard Gaussian vectors used by the finite run, including the initial state and all stochastic reinjections. Here, $N_G = 1$ for the purely deterministic sampler, $N_G = N + 1$ for the purely stochastic sampler, and $N_G = 1 + N_s$ for the hybrid sampler, where N_s is the number of stochastic steps. For $0 < \delta < 1$, set

$$r_\delta = \sqrt{d} + \sqrt{2 \log(N_G/\delta)}$$

and define the event

$$\Omega_\delta^{\text{loc}} = \left\{ \max_{1 \leq j \leq N_G} \|\Xi_j\|_2 \leq r_\delta \right\}.$$

The standard Gaussian norm tail bound and a union bound (Boucheron et al., 2013) imply

$$\mathbb{P}(\Omega_\delta^{\text{loc}}) \geq 1 - \delta.$$

Because the Gaussian initialization $\mathbf{x}_0$ is included among these vectors, set $R_0 = r_\delta$ and define recursively

$$\begin{aligned} R_{k+1} &= A_k R_k + B_k(1 + r_\delta) + \gamma_k G_c, \\ R_{k+1}^{\text{prop}} &= A_k R_k + B_k(1 + r_\delta). \end{aligned}$$

Then, on $\Omega_\delta^{\text{loc}}$, $\|\mathbf{x}_k\|_2 \leq R_k$ and $\|\mathbf{y}_{k+1}\|_2 \leq R_{k+1}^{\text{prop}}$. Let

$$R_\delta^{\text{all}} = \max_{0 \leq k < N} \{R_k, R_{k+1}^{\text{prop}}, R_{k+1}^{\text{prop}} + \gamma_k G_c\}, \quad \mathcal{K}_\delta = \overline{B}(0, R_\delta^{\text{all}}). \quad (32)$$

Here $\overline{B}(0, R)$ denotes the closed Euclidean ball of radius R . Because $\mathcal{K}_\delta$ is convex, it contains every state, every raw proposal, and every correction segment $[\mathbf{y}_{k+1}, \mathbf{x}_{k+1}]$ on $\Omega_\delta^{\text{loc}}$.

The remainder of the local analysis is stated on a fixed event $\Omega_\delta^{\text{loc}}$ and its associated compact set $\mathcal{K}_\delta$. For notational economy, we write $\mathcal{K} = \mathcal{K}_\delta$ and define

$$F_k(\mathbf{x}) = \mathcal{L}_h(E_{t_k}^\theta(\mathbf{x})), \quad F_k^* = \min_{\mathbf{x} \in \mathcal{K}} F_k(\mathbf{x}), \quad \Delta_k = F_k(\mathbf{x}_k) - F_k^*. \quad (33)$$

For stochastic steps, $\mathcal{F}_k$ denotes the history immediately before the fresh Gaussian vector at step k is drawn. All local constants may depend on the trained network, the fixed physical grid, the component weights, N , δ , and $\mathcal{K}$, but not on the step index. A statement that is uniform over a family of grids requires separate uniform control of these constants.

Assumption 2 (Local differentiability and endpoint regularity) *There is an open set $\mathcal{U} \supset \mathcal{K}$ containing all proposal and correction segments on which every F_k is continuously differentiable and*

$$\|\nabla F_k(\mathbf{x}) - \nabla F_k(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2, \quad \mathbf{x}, \mathbf{y} \in \mathcal{K},$$

for a common finite constant L . The endpoint maps E_t^θ are differentiable on the same open set.

The compactness of $\mathcal{K}$ is used only to delimit the region on which local derivatives and geometric constants are required. It does not imply the PL condition introduced later, and it is obtained independently of that condition.

3.2 Smoothness of the implemented MSE objective

The following results connect Assumption 2 to the component-normalized loss evaluated by the sampler.

Lemma 7 (One component-normalized MSE) *Let $\mathcal{Z}_0 \subset \mathbb{R}^m$ be convex and let $R : \mathcal{Z}_0 \rightarrow \mathbb{R}^r$ be continuously differentiable. Suppose*

$$\begin{aligned} \sup_{\mathbf{z} \in \mathcal{Z}_0} \|R(\mathbf{z})\|_2 &\leq B_0, \quad \sup_{\mathbf{z} \in \mathcal{Z}_0} \|J_R(\mathbf{z})\|_{\text{op}} \leq B_1, \\ \|J_R(\mathbf{z}) - J_R(\mathbf{z}')\|_{\text{op}} &\leq B_2 \|\mathbf{z} - \mathbf{z}'\|_2. \end{aligned}$$

For $Q(\mathbf{z}) = r^{-1} \|R(\mathbf{z})\|_2^2$,

$$\nabla Q(\mathbf{z}) = \frac{2}{r} J_R(\mathbf{z})^\top R(\mathbf{z}), \quad (34)$$

$$\|\nabla Q(\mathbf{z})\|_2 \leq \frac{2B_0B_1}{r}, \quad (35)$$

$$\|\nabla Q(\mathbf{z}) - \nabla Q(\mathbf{z}')\|_2 \leq \frac{2}{r} (B_1^2 + B_0B_2) \|\mathbf{z} - \mathbf{z}'\|_2. \quad (36)$$

If $R(\mathbf{z}) = A\mathbf{z} - b$ is affine, then $\nabla^2 Q = 2A^\top A/r$ and the exact global smoothness constant is $2\|A\|_{\text{op}}^2/r$.

Proof The chain rule gives Equation (34). Add and subtract $J_R(\mathbf{z})^\top R(\mathbf{z}')$ to obtain

$$\begin{aligned} \nabla Q(\mathbf{z}) - \nabla Q(\mathbf{z}') &= \frac{2}{r} J_R(\mathbf{z})^\top \{R(\mathbf{z}) - R(\mathbf{z}')\} \\ &\quad + \frac{2}{r} \{J_R(\mathbf{z}) - J_R(\mathbf{z}')\}^\top R(\mathbf{z}'). \end{aligned}$$

The integral mean-value identity gives

$$R(\mathbf{z}) - R(\mathbf{z}') = \int_0^1 J_R(\mathbf{z}' + s(\mathbf{z} - \mathbf{z}'))(\mathbf{z} - \mathbf{z}') \, ds,$$

so $\|R(\mathbf{z}) - R(\mathbf{z}')\|_2 \leq B_1 \|\mathbf{z} - \mathbf{z}'\|_2$. The two displayed terms are therefore bounded by $2B_1^2 \|\mathbf{z} - \mathbf{z}'\|_2/r$ and $2B_0B_2 \|\mathbf{z} - \mathbf{z}'\|_2/r$, respectively, proving Equation (36). Equation (35) follows directly from $\|J_R(\mathbf{z})^\top R(\mathbf{z})\|_2 \leq B_1B_0$.

If $R(\mathbf{z}) = A\mathbf{z} - b$, then

$$Q(\mathbf{z}) = \frac{1}{r} (A\mathbf{z} - b)^\top (A\mathbf{z} - b), \quad \nabla Q(\mathbf{z}) = \frac{2}{r} A^\top (A\mathbf{z} - b),$$

and a second differentiation gives $\nabla^2 Q(\mathbf{z}) = 2A^\top A/r$. Its operator norm is $2\|A\|_{\text{op}}^2/r$, which is the exact global gradient-Lipschitz constant. $\blacksquare$

Corollary 8 (Weighted physical objective) *Suppose Lemma 7 holds for every active component $j \in \mathcal{I}$ with constants $B_{0,j}, B_{1,j}, B_{2,j}$ and count m_j . Then, on $\mathcal{Z}_0$,*

$$\|\nabla \mathcal{L}_h(\mathbf{z})\|_2 \leq G_{\mathcal{L}} := \sum_{j \in \mathcal{I}} w_j \frac{2B_{0,j}B_{1,j}}{m_j}, \quad (37)$$

$$\|\nabla \mathcal{L}_h(\mathbf{z}) - \nabla \mathcal{L}_h(\mathbf{z}')\|_2 \leq L_{\mathcal{L}} \|\mathbf{z} - \mathbf{z}'\|_2, \quad (38)$$

$$L_{\mathcal{L}} := \sum_{j \in \mathcal{I}} w_j \frac{2(B_{1,j}^2 + B_{0,j}B_{2,j})}{m_j}. \quad (39)$$

For a coordinate-selection observation residual, let P_M be the matrix that selects the scalar entries indicated by the broadcast mask. Its exact component Hessian is $2P_M^\top P_M/m_j$.

Proof For each active component, define $Q_j(\mathbf{z}) = m_j^{-1} \|R_{h,j}(\mathbf{z})\|_2^2$. Since the active set is finite, Equation (7) may be differentiated term by term:

$$\nabla \mathcal{L}_h(\mathbf{z}) = \sum_{j \in \mathcal{I}} w_j \nabla Q_j(\mathbf{z}).$$

Equation (35) and the triangle inequality give

$$\begin{aligned} \|\nabla \mathcal{L}_h(\mathbf{z})\|_2 &\leq \sum_{j \in \mathcal{I}} w_j \|\nabla Q_j(\mathbf{z})\|_2 \\ &\leq \sum_{j \in \mathcal{I}} w_j \frac{2B_{0,j}B_{1,j}}{m_j} = G_{\mathcal{L}}. \end{aligned}$$

Similarly,

$$\begin{aligned} \|\nabla \mathcal{L}_h(\mathbf{z}) - \nabla \mathcal{L}_h(\mathbf{z}')\|_2 &\leq \sum_{j \in \mathcal{I}} w_j \|\nabla Q_j(\mathbf{z}) - \nabla Q_j(\mathbf{z}')\|_2 \\ &\leq \sum_{j \in \mathcal{I}} w_j \frac{2(B_{1,j}^2 + B_{0,j}B_{2,j})}{m_j} \|\mathbf{z} - \mathbf{z}'\|_2 \\ &= L_{\mathcal{L}} \|\mathbf{z} - \mathbf{z}'\|_2. \end{aligned}$$

For a coordinate-selection residual $R(\mathbf{z}) = P_M \mathbf{z} - \mathbf{y}$, the Jacobian is the constant matrix P_M . The affine part of Lemma 7 therefore gives the Hessian $2P_M^\top P_M/m_j$. $\blacksquare$

Corollary 8 is a local, fixed-grid regularity statement. The set $\mathcal{Z}_0$ should be understood as a convex subset of the decoded reachable endpoint region, not as the full infinite-dimensional PDE state space. The constants $B_{0,j}, B_{1,j}, B_{2,j}$ are therefore local constants evaluated after the finite-horizon localization region has been constructed.

For the observation components, the residuals are coordinate-selection maps of the form $R(\mathbf{z}) = P_M \mathbf{z} - \mathbf{y}$ and hence are affine. The same observation applies to Dirichlet, Neumann, open-boundary, periodic, and initial-condition residuals whenever they are implemented by fixed finite-difference or selection operators. These components satisfy the assumptions of

Lemma 7 on any bounded $\mathcal{Z}_0$, and their gradient-Lipschitz constants are exactly determined by the corresponding linear operators.

For the static elliptic equations, the Poisson and Helmholtz residuals are affine in the physical fields at fixed grid resolution and fixed wave number. Thus the affine part of Lemma 7 applies directly. The Darcy residual $\nabla \cdot (a\nabla p) + 1$ is bilinear in the coefficient and pressure variables after discretization; it is therefore smooth on bounded subsets of the physical endpoint space, but the constants are local and need not remain bounded on the whole space. The binary thresholding used to generate Darcy coefficients is part of the data-generation distribution, not part of the differentiated residual map.

For the temporal endpoint equations with fixed scalar metadata, the Heat, first-order Wave, and Advection-Diffusion residual maps are linear in the state variables at fixed grid resolution. If the scalar parameters α , c , b_x , b_y , or κ are treated as additional unknowns rather than fixed metadata, the residuals become at most bilinear in the state-parameter pair and still satisfy the lemma locally on bounded sets. The Burgers residual based on the stored full space-time field is polynomial in the field variables because of the term $u\partial_x u$; hence it is locally smooth on bounded subsets but not globally gradient-bounded.

For the nonlinear temporal residuals, the Reaction-Diffusion residual is polynomial in (u, v) and therefore satisfies the lemma on bounded reachable sets, although its Jacobian is not globally bounded because of the cubic reaction term. In the periodic vorticity formulation of Navier-Stokes, the FFT differentiation, zero-mean projection, and stream-function solve are fixed linear maps at a fixed resolution, while the transport term $\mathbf{v} \cdot \nabla \omega$ is quadratic in vorticity. Consequently, the finite-dimensional residual is smooth on bounded vorticity sets, provided the zero mode is handled consistently. The associated constants may grow with spatial resolution, high-frequency content, decreasing viscosity, or longer physical horizons.

The main cases where the assumptions may fail are the residuals with hard nonsmooth safeguards or singular physical variables. For Steady Heat Conduction, the conductivity $\lambda(u) = \max\{1 + 0.05(u - 298), 0.1\}$ is only piecewise smooth; the stated regularity holds on reachable sets that remain away from the switching surface, or if the hard maximum is replaced by a smooth positive parameterization. For Shallow Water, the fluxes contain divisions by the water depth and the implementation may use a safe-depth clamp such as $h_\varepsilon = \max(h, \varepsilon_h)$. The assumptions therefore hold only on wet compact regions separated from dry states and from the clamp switching surface. If the trajectory reaches $h = 0$, the clamp threshold, or another nonsmooth safeguard, Lemma 7 and Corollary 8 should be interpreted as inapplicable unless a smooth regularization is used.

Thus the corollary covers all affine and linear residual components directly and covers the polynomial or bilinear nonlinear residuals locally on bounded fixed-grid regions.

Lemma 9 (Endpoint-map and composition constants) *The sampler applies channel-wise inverse standardization and then a fixed channel split. After vectorization, this physical-coordinate decoder has the affine form*

$$\mathcal{D}_{\text{phys}}(\mathbf{y}) = S\mathbf{y} + o_{\text{phys}}, \quad (40)$$

where S is the fixed linear operator formed by channel-wise scaling and the channel split, and o_{phys} is the fixed offset formed by the channel-wise means. For the usual paired states,

S is block diagonal up to channel selection. For the single-channel Burgers representation, the fixed split may duplicate the same physical channel. The argument requires only that S is constant. Hence, for any differentiable endpoint predictor P_t^θ ,

$$J_{E_t^\theta}(\mathbf{x}) = SJ_{P_t^\theta}(\mathbf{x}). \quad (41)$$

For the Euler CondOT predictor in Equation (3), this becomes

$$J_{E_t^\theta}(\mathbf{x}) = S\{I + (1-t)J_{v_t^\theta}(\mathbf{x})\}. \quad (42)$$

Let $\mathcal{Z}_\mathcal{K}^{\text{reg}}$ be a convex set containing $E_t^\theta(\mathcal{K})$ for all analyzed flow times. Suppose that, for $\mathbf{x}, \mathbf{y} \in \mathcal{K}$ and analyzed s, t ,

$$\|J_{E_t^\theta}(\mathbf{x})\|_{\text{op}} \leq M_E, \quad (43)$$

$$\|J_{E_t^\theta}(\mathbf{x}) - J_{E_t^\theta}(\mathbf{y})\|_{\text{op}} \leq H_E\|\mathbf{x} - \mathbf{y}\|_2, \quad (44)$$

$$\|E_t^\theta(\mathbf{x}) - E_s^\theta(\mathbf{x})\|_2 \leq M_{E,t}|t - s|. \quad (45)$$

Suppose also that Corollary 8 holds on $\mathcal{Z}_\mathcal{K}^{\text{reg}}$. Then the composite objective $F_t = \mathcal{L}_h \circ E_t^\theta$ satisfies

$$\|\nabla F_t(\mathbf{x}) - \nabla F_t(\mathbf{y})\|_2 \leq L_F\|\mathbf{x} - \mathbf{y}\|_2, \quad L_F = G_\mathcal{L}H_E + L_\mathcal{L}M_E^2, \quad (46)$$

$$|F_t(\mathbf{x}) - F_s(\mathbf{x})| \leq G_\mathcal{L}M_{E,t}|t - s|. \quad (47)$$

In particular, Assumption 2 holds with $L = L_F$.

Proof Equation (41) follows because the Jacobian of the affine map in Equation (40) is the constant operator S . Differentiating $P_t^\theta(\mathbf{x}) = \mathbf{x} + (1-t)v_t^\theta(\mathbf{x})$ proves Equation (42).

Define the physical endpoint gradient

$$\psi_t(\mathbf{x}) = \nabla_z \mathcal{L}_h(E_t^\theta(\mathbf{x})).$$

The chain rule gives

$$\nabla F_t(\mathbf{x}) = J_{E_t^\theta}(\mathbf{x})^\top \psi_t(\mathbf{x}).$$

Adding and subtracting $J_{E_t^\theta}(\mathbf{y})^\top \psi_t(\mathbf{x})$ gives

$$\nabla F_t(\mathbf{x}) - \nabla F_t(\mathbf{y}) = \{J_{E_t^\theta}(\mathbf{x}) - J_{E_t^\theta}(\mathbf{y})\}^\top \psi_t(\mathbf{x}) + J_{E_t^\theta}(\mathbf{y})^\top \{\psi_t(\mathbf{x}) - \psi_t(\mathbf{y})\}.$$

The first term is bounded by $H_E G_\mathcal{L}\|\mathbf{x} - \mathbf{y}\|_2$. For the second term, Corollary 8 and Equation (43) imply

$$\|\psi_t(\mathbf{x}) - \psi_t(\mathbf{y})\|_2 \leq L_\mathcal{L}\|E_t^\theta(\mathbf{x}) - E_t^\theta(\mathbf{y})\|_2 \leq L_\mathcal{L}M_E\|\mathbf{x} - \mathbf{y}\|_2,$$

so the second term is at most $L_\mathcal{L}M_E^2\|\mathbf{x} - \mathbf{y}\|_2$. This proves Equation (46).

Finally, the gradient bound in Equation (37) makes $\mathcal{L}_h$ $G_\mathcal{L}$ -Lipschitz along line segments in $\mathcal{Z}_\mathcal{K}^{\text{reg}}$. Therefore

$$|F_t(\mathbf{x}) - F_s(\mathbf{x})| \leq G_\mathcal{L}\|E_t^\theta(\mathbf{x}) - E_s^\theta(\mathbf{x})\|_2 \leq G_\mathcal{L}M_{E,t}|t - s|,$$

which proves Equation (47). ■

The affine form in Equation (40) is an exact property of the implemented inverse-standardization map. The nonlinear part of E_t^θ is entirely in P_t^θ . In particular, if $(t, \mathbf{x}) \mapsto P_t^\theta(\mathbf{x})$ is twice continuously differentiable on an open neighborhood of the compact reachable region, then the finite constants M_E , H_E , and $M_{E,t}$ required above exist automatically on that region by continuity. No lower singular-value or global invertibility condition is implied.

3.3 Local PL geometry and non-identifiability

The Polyak-Łojasiewicz (PL) inequality converts a gradient-norm decrease into objective-value contraction without requiring convexity (Karimi et al., 2016).

Assumption 3 (Local Polyak-Łojasiewicz condition) *There is $\mu > 0$ such that, for every analyzed k and every $\mathbf{x} \in \mathcal{K}$,*

$$\frac{1}{2} \|\nabla F_k(\mathbf{x})\|_2^2 \geq \mu \{F_k(\mathbf{x}) - F_k^*\}. \quad (48)$$

This condition is used only to convert descent or stationarity into objective contraction; it is not used for finite-horizon stability.

Proposition 10 (Compatible residual-map condition) *Suppose $F_k(\mathbf{x}) = \frac{1}{2} \|r_k(\mathbf{x})\|_2^2$, $F_k^* = 0$, and let r_k collect the active residuals. For the implemented composite objective, one may take $r_k(\mathbf{x})$ to be the stacked vector with blocks $\sqrt{2w_j/m_j} R_{h,j}(E_{t_k}^\theta(\mathbf{x}))$, $j \in \mathcal{I}$. Assume*

$$\|J_{r_k}(\mathbf{x})^\top r_k(\mathbf{x})\|_2 \geq \sigma \|r_k(\mathbf{x})\|_2 \quad (49)$$

for some $\sigma > 0$ and all $\mathbf{x} \in \mathcal{K}$. Then Assumption 3 holds with $\mu = \sigma^2$.

Proof The chain rule for the squared residual gives

$$\nabla F_k(\mathbf{x}) = J_{r_k}(\mathbf{x})^\top r_k(\mathbf{x}).$$

Applying Equation (49) and then squaring both sides yields

$$\|\nabla F_k(\mathbf{x})\|_2^2 \geq \sigma^2 \|r_k(\mathbf{x})\|_2^2.$$

Because $F_k(\mathbf{x}) = \|r_k(\mathbf{x})\|_2^2/2$ and $F_k^* = 0$,

$$\frac{1}{2} \|\nabla F_k(\mathbf{x})\|_2^2 \geq \frac{\sigma^2}{2} \|r_k(\mathbf{x})\|_2^2 = \sigma^2 \{F_k(\mathbf{x}) - F_k^*\}.$$

This is Equation (48) with $\mu = \sigma^2$. ■

Proposition 11 (Affine residuals and a set of minimizers) *Let $F(\mathbf{x}) = \frac{1}{2}\|A\mathbf{x} - b\|_2^2$, let $F^\star = \min_{\mathbf{x}} F(\mathbf{x})$, let $\sigma_+(A) > 0$ be the smallest positive singular value of A , and let $\mathcal{X}^\star = \arg \min_{\mathbf{x}} F(\mathbf{x})$. For a nonempty set $\mathcal{A}$, write $\text{dist}(\mathbf{x}, \mathcal{A}) = \inf_{\mathbf{y} \in \mathcal{A}} \|\mathbf{x} - \mathbf{y}\|_2$. Then*

$$\frac{1}{2}\|\nabla F(\mathbf{x})\|_2^2 \geq \sigma_+(A)^2\{F(\mathbf{x}) - F^\star\}, \quad (50)$$

$$\text{dist}(\mathbf{x}, \mathcal{X}^\star)^2 \leq \frac{2}{\sigma_+(A)^2}\{F(\mathbf{x}) - F^\star\}. \quad (51)$$

The minimizer need not be unique when A has a nullspace.

Proof Let $\mathbf{x}^\star \in \mathcal{X}^\star$ be any least-squares minimizer and put $d = \mathbf{x} - \mathbf{x}^\star$. The normal equations give $A^\top(A\mathbf{x}^\star - b) = 0$. Expanding the square and using this orthogonality,

$$\begin{aligned} F(\mathbf{x}) &= \frac{1}{2}\|A\mathbf{x}^\star - b + Ad\|_2^2 \\ &= \frac{1}{2}\|A\mathbf{x}^\star - b\|_2^2 + \langle A\mathbf{x}^\star - b, Ad \rangle + \frac{1}{2}\|Ad\|_2^2 \\ &= F^\star + \frac{1}{2}\|Ad\|_2^2. \end{aligned}$$

Moreover,

$$\nabla F(\mathbf{x}) = A^\top(A\mathbf{x} - b) = A^\top Ad.$$

Let $A = U\Sigma V^\top$ be a singular-value decomposition and define

$$V^\top d = (c_1, \dots, c_n)^\top.$$

Summing only over the positive singular values gives

$$\begin{aligned} 2\{F(\mathbf{x}) - F^\star\} &= \|Ad\|_2^2 = \sum_{i:\sigma_i>0} \sigma_i^2 c_i^2, \\ \|\nabla F(\mathbf{x})\|_2^2 &= \|A^\top Ad\|_2^2 = \sum_{i:\sigma_i>0} \sigma_i^4 c_i^2. \end{aligned}$$

Since $\sigma_i \geq \sigma_+(A)$ on every nonzero singular component,

$$\|\nabla F(\mathbf{x})\|_2^2 \geq \sigma_+(A)^2\|Ad\|_2^2 = 2\sigma_+(A)^2\{F(\mathbf{x}) - F^\star\},$$

which proves Equation (50).

The minimizer set is the affine space $\mathcal{X}^\star = \mathbf{x}^\star + \ker(A)$. If $d = d_0 + d_\perp$ with $d_0 \in \ker(A)$ and $d_\perp \in \ker(A)^\perp$, then

$$\text{dist}(\mathbf{x}, \mathcal{X}^\star) = \|d_\perp\|_2$$

and $Ad = Ad_\perp$. The definition of $\sigma_+(A)$ gives

$$\|Ad\|_2 \geq \sigma_+(A)\|d_\perp\|_2.$$

Combining this inequality with $2\{F(\mathbf{x}) - F^\star\} = \|Ad\|_2^2$ proves Equation (51). ■

Proposition 11 applies directly to affine residuals in physical endpoint coordinates, including masked observation operators. After composition with a nonlinear endpoint predictor, the resulting model-space objective need not remain affine; its PL geometry must then be checked through Proposition 10, local strong convexity, or another problem-specific error bound.

Proposition 12 (Local strong convexity) *Let $\mathcal{V} \subseteq \mathcal{K}$ be convex. Suppose F_k is twice continuously differentiable on an open convex neighborhood of $\mathcal{V}$, there is $\mathbf{x}_k^* \in \mathcal{V}$ with $F_k(\mathbf{x}_k^*) = F_k^*$, and $\nabla^2 F_k(\mathbf{x}) \succeq mI$ for every $\mathbf{x} \in \mathcal{V}$ and some $m > 0$, where $\succeq$ denotes the positive-semidefinite order. Then Equation (48) holds on $\mathcal{V}$ with $\mu = m$, including when $F_k^* > 0$.*

Proof The Hessian lower bound on the convex set $\mathcal{V}$ implies the strong convexity inequality

$$F_k(\mathbf{y}) \geq F_k(\mathbf{x}) + \langle \nabla F_k(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|_2^2, \quad \mathbf{x}, \mathbf{y} \in \mathcal{V}.$$

Set $\mathbf{y} = \mathbf{x}_k^*$ and rearrange:

$$F_k(\mathbf{x}) - F_k(\mathbf{x}_k^*) \leq \langle \nabla F_k(\mathbf{x}), \mathbf{x} - \mathbf{x}_k^* \rangle - \frac{m}{2} \|\mathbf{x} - \mathbf{x}_k^*\|_2^2.$$

With $r = \|\mathbf{x} - \mathbf{x}_k^*\|_2$, Cauchy-Schwarz bounds the right-hand side by

$$\|\nabla F_k(\mathbf{x})\|_2 r - \frac{m}{2} r^2.$$

The scalar quadratic attains its maximum over $r \geq 0$ at $r = \|\nabla F_k(\mathbf{x})\|_2 / m$, with maximum $\|\nabla F_k(\mathbf{x})\|_2^2 / (2m)$. Hence

$$F_k(\mathbf{x}) - F_k^* \leq \frac{1}{2m} \|\nabla F_k(\mathbf{x})\|_2^2,$$

which is the PL inequality with $\mu = m$. The value $F_k(\mathbf{x}_k^*)$ need not be zero. ■

These propositions describe local sufficient conditions, not universal properties of the PDE experiments. Sparse masks leave unobserved directions, Helmholtz inverse problems may be ill-conditioned or nonunique, and nonlinear residuals may possess symmetries or nearly flat directions. The reported theory therefore remains conditional even when an objective-value PL inequality holds for a local basin or for a set-valued least-squares problem.

3.4 Effective guidance under active clipping

Finite-horizon localization controls the magnitude of the applied correction without assuming that clipping is inactive. Once the compact analysis region has been constructed, compactness also provides a positive lower bound on the scalar clipping factor. This lower bound is used only in results that require a nonvanishing cumulative guidance budget.

Lemma 13 (Local lower bound for the clipping factor) *Let*

$$G_{\mathcal{K}} = \max_{0 \leq k < N} \max_{\mathbf{x} \in \mathcal{K}} \|\nabla F_k(\mathbf{x})\|_2. \tag{52}$$

Under Assumption 2, $G_{\mathcal{K}} < \infty$. Define

$$\rho_{\mathcal{K}} = \min\left\{1, \frac{G_c}{G_{\mathcal{K}} + \varepsilon_{\text{clip}}}\right\} > 0.$$

Then, on the localization event $\Omega_{\delta}^{\text{loc}}$, $\rho_k \geq \rho_{\mathcal{K}}$ for every analyzed step. Consequently, on any phase for which $\gamma_k \geq \underline{\gamma} > 0$,

$$a_k = \gamma_k \rho_k \geq \underline{\gamma} \rho_{\mathcal{K}} > 0. \quad (53)$$

Proof The finitely many gradients ∇F_k are continuous on the compact set $\mathcal{K}$, so the maximum in Equation (52) is finite. Hence

$$\|g_k\|_2 \leq G_{\mathcal{K}}$$

on $\Omega_{\delta}^{\text{loc}}$. The function $r \mapsto \min\{1, G_c/(r + \varepsilon_{\text{clip}})\}$ is nonincreasing for $r \geq 0$, and substitution into Equation (26) proves $\rho_k \geq \rho_{\mathcal{K}}$. Multiplying by the lower bound for γ_k proves Equation (53). $\blacksquare$

The order of the argument is important: the radius of $\mathcal{K}$ follows from the bounded *applied* correction in Lemma 5 and Theorem 6; only afterwards is compactness used to lower-bound ρ_k . Thus active clipping is retained explicitly and no invariant region is inferred from an unclipped recursion.

3.5 Endpoint-transport defects

The raw proposal changes both flow time and state before the guidance correction is applied. It is therefore useful to distinguish the gradient computed at the pre-proposal pair $(t_k, \mathbf{x}_k)$ from the gradient associated with the post-proposal pair $(t_{k+1}, \mathbf{y}_{k+1})$. Define

$$\mathbf{y}_{k+1} = \mathcal{A}_k(\mathbf{x}_k, \xi_k), \quad g_k = g_k^- := \nabla F_k(\mathbf{x}_k), \quad g_k^+ := \nabla F_{k+1}(\mathbf{y}_{k+1}).$$

The superscripts $-$ and $+$ refer only to the two sides of the raw proposal; they do not denote old and new network parameters. The implemented step uses g_k^- to correct $\mathbf{y}_{k+1}$. Its mismatch with the gradient naturally associated with the proposal is

$$e_k = g_k^- - g_k^+. \quad (54)$$

We also define the nonnegative objective-transport defect, using $[r]_+ = \max\{r, 0\}$,

$$p_k = [F_{k+1}(\mathbf{y}_{k+1}) - F_{k+1}^* - \{F_k(\mathbf{x}_k) - F_k^*\}]_+.$$

Thus e_k records the stale-gradient error created by moving in flow time and state, whereas p_k records only the harmful increase in suboptimality caused by the unguided proposal. The raw objective change and e_k can be evaluated from a saved trace by one additional endpoint evaluation and backward pass at $(t_{k+1}, \mathbf{y}_{k+1})$. Evaluating p_k itself also requires estimates of the local minima F_k^* and F_{k+1}^* ; the next lemma provides bounds that separate this minimum drift from directly measurable endpoint transport.

Lemma 14 (Bounds for the endpoint-transport defects) *Assume on the relevant de-coded endpoint region that*

$$\begin{aligned} \|\nabla \mathcal{L}_h(\mathbf{z})\|_2 &\leq G_{\mathcal{L}}, \quad \|J_{E_t^\theta}(\mathbf{x})\|_{\text{op}} \leq M_E, \\ \|\nabla \mathcal{L}_h(\mathbf{z}) - \nabla \mathcal{L}_h(\mathbf{z}')\|_2 &\leq L_{\mathcal{L}}\|\mathbf{z} - \mathbf{z}'\|_2. \end{aligned}$$

Set

$$\begin{aligned} \varepsilon_{0,k} &= \|E_{t_{k+1}}^\theta(\mathbf{y}_{k+1}) - E_{t_k}^\theta(\mathbf{x}_k)\|_2, \\ \varepsilon_{1,k} &= \|J_{E_{t_{k+1}}^\theta}(\mathbf{y}_{k+1}) - J_{E_{t_k}^\theta}(\mathbf{x}_k)\|_{\text{op}}, \\ \varepsilon_{\star,k} &= |F_{k+1}^\star - F_k^\star|. \end{aligned}$$

Then

$$p_k \leq G_{\mathcal{L}}\varepsilon_{0,k} + \varepsilon_{\star,k}, \quad (55)$$

$$\|e_k\|_2 \leq G_{\mathcal{L}}\varepsilon_{1,k} + M_E L_{\mathcal{L}}\varepsilon_{0,k}. \quad (56)$$

If $\sup_{\mathbf{x} \in \mathcal{K}} \|E_{t_{k+1}}^\theta(\mathbf{x}) - E_{t_k}^\theta(\mathbf{x})\|_2 \leq \bar{\varepsilon}_{t,k}$, then $\varepsilon_{\star,k} \leq G_{\mathcal{L}}\bar{\varepsilon}_{t,k}$.

Proof The gradient bound makes $\mathcal{L}_h$ locally $G_{\mathcal{L}}$ -Lipschitz, hence

$$|F_{k+1}(\mathbf{y}_{k+1}) - F_k(\mathbf{x}_k)| \leq G_{\mathcal{L}}\varepsilon_{0,k}.$$

Adding and subtracting the two minimum values proves Equation (55). Let E_+, E_0 and J_+, J_0 denote the two endpoint values and Jacobians. Then

$$\begin{aligned} \|e_k\|_2 &= \|J_0^\top \nabla \mathcal{L}_h(E_0) - J_+^\top \nabla \mathcal{L}_h(E_+)\|_2 \\ &\leq G_{\mathcal{L}}\|J_0 - J_+\|_{\text{op}} + M_E L_{\mathcal{L}}\|E_0 - E_+\|_2, \end{aligned}$$

which proves Equation (56). The final statement follows from $|\min f - \min g| \leq \sup |f - g|$ and the same Lipschitz bound. $\blacksquare$

Proposition 15 (Deterministic transport expansion) *Assume that $(t, \mathbf{x}) \mapsto E_t^\theta(\mathbf{x})$ is twice continuously differentiable on an open neighborhood containing the deterministic Euler proposal segments. Define the endpoint convective residual*

$$\mathcal{R}_E^\theta(t, \mathbf{x}) = \partial_t E_t^\theta(\mathbf{x}) + J_{E_t^\theta}(\mathbf{x})v_t^\theta(\mathbf{x}).$$

Suppose $\|\mathcal{R}_E^\theta(t, \mathbf{x})\|_2 \leq \epsilon_E$ on this neighborhood and that the first and second derivatives of E_t^θ are bounded there. Then finite constants C_0, C_1 exist such that the deterministic Euler proposal satisfies

$$\varepsilon_{0,k} \leq \Delta t_k \epsilon_E + C_0(\Delta t_k)^2, \quad \varepsilon_{1,k} \leq C_1 \Delta t_k.$$

For comparison, let $\varphi_{t,s}$ be the characteristic flow generated by an exact velocity field and define its terminal-flow map by $E_t^{\text{flow}}(\mathbf{x}) = \varphi_{t,1}(\mathbf{x})$. The semigroup property implies

$$\partial_t E_t^{\text{flow}}(\mathbf{x}) + J_{E_t^{\text{flow}}}(\mathbf{x})v_t(\mathbf{x}) = 0. \quad (57)$$

The CondOT predictor $\mathbf{x} + (1-t)v_t(\mathbf{x})$ is instead a conditional-mean endpoint predictor and need not satisfy Equation (57). Accordingly, $\mathcal{R}_E^\theta$ is retained as an explicit transport defect rather than assumed to vanish.

Proof Write $\Delta t = \Delta t_k$, $\mathbf{x} = \mathbf{x}_k$, and $v = v_{t_k}^\theta(\mathbf{x}_k)$. Define

$$\phi(r) = E_{t_k+r}^\theta(\mathbf{x} + rv), \quad 0 \leq r \leq \Delta t.$$

The deterministic Euler proposal is $\mathbf{y}_{k+1} = \mathbf{x} + \Delta t v$, so

$$\varepsilon_{0,k} = \|\phi(\Delta t) - \phi(0)\|_2.$$

By the chain rule,

$$\phi'(0) = \partial_t E_{t_k}^\theta(\mathbf{x}) + J_{E_{t_k}^\theta}(\mathbf{x}) v_{t_k}^\theta(\mathbf{x}) = \mathcal{R}_E^\theta(t_k, \mathbf{x}).$$

The stated derivative bounds imply that ϕ' is locally Lipschitz along the segment; hence, after choosing a finite C_0 ,

$$\|\phi'(r) - \phi'(0)\|_2 \leq 2C_0 r, \quad 0 \leq r \leq \Delta t.$$

The fundamental theorem of calculus therefore gives

$$\begin{aligned} \|\phi(\Delta t) - \phi(0)\|_2 &\leq \Delta t \|\phi'(0)\|_2 + \int_0^{\Delta t} \|\phi'(r) - \phi'(0)\|_2 \, dr \\ &\leq \Delta t \epsilon_E + C_0 (\Delta t)^2, \end{aligned}$$

which proves the first estimate.

For the Jacobian defect, apply the mean-value theorem to the continuously differentiable map $(t, \mathbf{x}) \mapsto J_{E_t^\theta}(\mathbf{x})$ along the same segment. The bounded derivative of this map gives

$$\|J_{E_{t_k+\Delta t}^\theta}(\mathbf{x} + \Delta t v) - J_{E_{t_k}^\theta}(\mathbf{x})\|_{\text{op}} \leq C_1 \Delta t,$$

proving the second estimate.

Finally, if $X_s = \varphi_{t,s}(\mathbf{x})$, then $E_s^{\text{flow}}(X_s) = \varphi_{s,1}(\varphi_{t,s}(\mathbf{x})) = \varphi_{t,1}(\mathbf{x})$ is constant in s . Differentiating this identity at $s = t$ proves Equation (57). $\blacksquare$

3.6 One-step descent and the unified finite-horizon bound

Lemma 16 (One-step inexact endpoint descent) *Suppose Assumption 2 holds and the complete step lies in $\mathcal{K}$. If*

$$0 \leq a_k \leq \frac{1}{2L},$$

where the upper restriction is interpreted as vacuous when $L = 0$, then the implemented update satisfies

$$F_{k+1}(\mathbf{x}_{k+1}) - F_{k+1}^\star \leq \Delta_k - \frac{a_k}{2} \|g_k\|_2^2 + p_k + a_k \|e_k\|_2^2. \quad (58)$$

If Assumption 3 also holds, then

$$\Delta_{k+1} \leq (1 - \mu a_k) \Delta_k + p_k + a_k \|e_k\|_2^2. \quad (59)$$

Proof The correction segment is contained in $\mathcal{K}$, so the descent lemma for the L -smooth function F_{k+1} may be applied to $\mathbf{x}_{k+1} = \mathbf{y}_{k+1} - a_k g_k$:

$$F_{k+1}(\mathbf{x}_{k+1}) \leq F_{k+1}(\mathbf{y}_{k+1}) - a_k \langle g_k^+, g_k \rangle + \frac{La_k^2}{2} \|g_k\|_2^2. \quad (60)$$

By Equation (54), $g_k^+ = g_k - e_k$. Hence the two correction terms on the right-hand side of Equation (60) are

$$-a_k \|g_k\|_2^2 + a_k \langle e_k, g_k \rangle + \frac{La_k^2}{2} \|g_k\|_2^2 = -a_k \left(1 - \frac{La_k}{2}\right) \|g_k\|_2^2 + a_k \langle e_k, g_k \rangle.$$

Young's inequality gives

$$\langle e_k, g_k \rangle \leq \|e_k\|_2^2 + \frac{1}{4} \|g_k\|_2^2.$$

Because $a_k \leq 1/(2L)$, we have $1 - La_k/2 \geq 3/4$. Consequently,

$$F_{k+1}(\mathbf{x}_{k+1}) \leq F_{k+1}(\mathbf{y}_{k+1}) - \frac{a_k}{2} \|g_k\|_2^2 + a_k \|e_k\|_2^2. \quad (61)$$

The definition of the nonnegative objective defect implies

$$F_{k+1}(\mathbf{y}_{k+1}) - F_{k+1}^* \leq \Delta_k + p_k.$$

Subtracting F_{k+1}^* in Equation (61) proves Equation (58). Under the local PL condition, $\|g_k\|_2^2 \geq 2\mu\Delta_k$; substitution gives Equation (59). $\blacksquare$

Corollary 17 (Stationarity without PL) *Under the first part of Lemma 16, let $d_k = p_k + a_k \|e_k\|_2^2$. Then*

$$\frac{1}{2} \sum_{k=0}^{N-1} a_k \|g_k\|_2^2 \leq \Delta_0 + \sum_{k=0}^{N-1} d_k.$$

Consequently, if $A_N = \sum_{k=0}^{N-1} a_k > 0$,

$$\min_{0 \leq k < N} \|g_k\|_2^2 \leq \frac{2}{A_N} \left(\Delta_0 + \sum_{k=0}^{N-1} d_k \right).$$

Proof Rearrange Equation (58) and sum. The objective gaps telescope and $\Delta_N \geq 0$. The minimum-gradient bound follows by lower bounding the weighted sum by $A_N \min_{k < N} \|g_k\|_2^2$. The final statement follows by contradiction. $\blacksquare$

Theorem 18 (Unified finite-horizon convergence) *Suppose Assumption 3 holds and Lemma 16 applies for $k = 0, 1, \dots, N-1$, with $0 \leq \mu a_k < 1$. Define*

$$\Gamma_{r,s} = \prod_{i=r}^{s-1} (1 - \mu a_i), \quad \Gamma_{s,s} = 1.$$

Then, pathwise on the localization event,

$$\Delta_N \leq \Gamma_{0,N} \Delta_0 + \sum_{j=0}^{N-1} \Gamma_{j+1,N} \{p_j + a_j \|e_j\|_2^2\}. \quad (62)$$

Proof Let $d_k = p_k + a_k \|e_k\|_2^2$. We prove the bound for every terminal index $m = 1, \dots, N$. For $m = 1$, the recursion (59) gives

$$\Delta_1 \leq (1 - \mu a_0) \Delta_0 + d_0 = \Gamma_{0,1} \Delta_0 + \Gamma_{1,1} d_0.$$

Assume the bound holds for some $m < N$. Applying (59) at step m and using the induction hypothesis,

$$\begin{aligned} \Delta_{m+1} &\leq (1 - \mu a_m) \Delta_m + d_m \\ &\leq (1 - \mu a_m) \left\{ \Gamma_{0,m} \Delta_0 + \sum_{j=0}^{m-1} \Gamma_{j+1,m} d_j \right\} + d_m \\ &= \Gamma_{0,m+1} \Delta_0 + \sum_{j=0}^{m-1} \Gamma_{j+1,m+1} d_j + \Gamma_{m+1,m+1} d_m. \end{aligned}$$

Thus the bound holds for $m + 1$. Taking $m = N$ proves (62). $\blacksquare$

The first term in Equation (62) is the optimization transient. The weighted sum is the complete proposal and endpoint-transport floor. Arbitrary accuracy follows only when the effective guidance budget grows and this weighted defect vanishes.

Corollary 19 (Transparent grid-refinement bound) *Suppose, on a phase of flow-time length $T_f = \sum_k \Delta t_k$, that for constants $c_a, C_a, C_p, C_e > 0$ and $r \geq 1$,*

$$c_a \Delta t_k \leq a_k \leq C_a \Delta t_k, \quad p_k \leq C_p (\Delta t_k)^{1+r}, \quad \|e_k\|_2 \leq C_e (\Delta t_k)^r. \quad (63)$$

With $H = \max_k \Delta t_k$,

$$\Delta_N \leq e^{-\mu c_a T_f} \Delta_0 + T_f \{C_p H^r + C_a C_e^2 H^{2r}\}.$$

Proof Use $\Gamma_{0,N} \leq \exp(-\mu \sum_k a_k)$ in Equation (62), discard the remaining factors bounded by one, and apply $\sum_k (\Delta t_k)^{1+r} \leq H^r T_f$ and $\sum_k (\Delta t_k)^{1+2r} \leq H^{2r} T_f$. $\blacksquare$

Refining a grid removes discretization and transport defects only under the explicit scaling in Equation (63); it does not by itself guarantee a growing guidance budget.

3.7 Deterministic contraction above an explicit floor

The next result is a phase-wise simplification of Theorem 18. The constant B_D is not an additional structural PDE assumption; it summarizes the deterministic proposal and endpoint-transport defects on the localized phase. It can be bounded by Lemma 14.

Theorem 20 (Deterministic phase-wise contraction) *Consider K deterministic corrections on the localization event. Suppose*

$$0 < a_D \leq a_k \leq \frac{1}{2L}, \quad 0 < \mu a_k < 1, \quad \|p_k + a_k\|_2^2 \leq B_D.$$

Then

$$\Delta_K \leq e^{-\mu a_D K} \Delta_0 + \frac{B_D}{\mu a_D} =: e^{-\mu a_D K} \Delta_0 + \varepsilon_D^{\text{floor}}.$$

For a requested accuracy $\varepsilon^ > \varepsilon_D^{\text{floor}}$, it is sufficient that*

$$K \geq \frac{1}{\mu a_D} \log \left(\frac{\Delta_0}{\varepsilon^* - \varepsilon_D^{\text{floor}}} \right) \quad (64)$$

whenever the logarithm is positive. Thus logarithmic iteration complexity is claimed only for targets strictly above the displayed floor. If $B_D = 0$, the same expression covers any positive target.

Proof Theorem 18 and the uniform bounds give

$$\Delta_K \leq (1 - \mu a_D)^K \Delta_0 + B_D \sum_{j=0}^{K-1} (1 - \mu a_D)^j.$$

Use $(1 - z)^K \leq e^{-zK}$ and bound the geometric sum by $1/(\mu a_D)$. Solving the transient inequality gives Equation (64). $\blacksquare$

The lower bound on a_k is compatible with active clipping: on a compact reachable set it follows from Lemma 13 whenever the raw deterministic multiplier is bounded below.

Corollary 21 (CondOT geometric deterministic phase) *Treat the numerical step from $t_0 = 0$ separately. Suppose subsequent deterministic points satisfy $t_{k+1} = (1 + \eta_{\text{grid}})t_k$ for a fixed $\eta_{\text{grid}} > 0$ and every analyzed point in the phase satisfies $t_k \leq t_{\text{sw}} < 1$. Then*

$$\gamma_k^D = \Delta t_k b_{t_k}^{\text{fm}} = \eta_{\text{grid}}(1 - t_k) \geq \eta_{\text{grid}}(1 - t_{\text{sw}}), \quad (65)$$

and

$$a_k \geq a_D := \eta_{\text{grid}}(1 - t_{\text{sw}})\rho_K. \quad (66)$$

If the upper step restriction and defect bound in Theorem 20 also hold, its floor-aware conclusion applies to this phase.

Proof For CondOT, $b_t^{\text{fm}} = (1 - t)/t$ and $\Delta t_k = \eta_{\text{grid}} t_k$, proving Equation (65). Apply Lemma 13 to obtain Equation (66). $\blacksquare$

This corollary explains why a geometric grid can regularize the product $\Delta t_k b_{t_k}^{\text{fm}}$ near zero. The numerically clamped zero-time correction is outside this geometric-grid identity and is included only when its realized effective step satisfies the descent condition; otherwise it is treated as initialization.

3.8 Expected stochastic contraction under localization

For the default stochastic branch, the effective guidance step is

$$a_k = c_\zeta(1 - t_k)\rho_k.$$

Fresh Gaussian reinjection affects the transport defects p_k and e_k . The coefficient of the newly drawn Gaussian vector in the final raw proposal is nevertheless zero because $t_N = 1$.

The next theorem is stated under a probability law supported on the localized run. This avoids using local smoothness or local PL conditions on trajectories that have left their domain of validity. The law may be the law of the process stopped at its first exit, a bounded-noise implementation, or the conditional law given the event $\Omega_\delta^{\text{loc}}$.

Theorem 22 (Expected stochastic endpoint bound) *Let $\mathbb{Q}$ be a probability law under which Assumption 3 and the hypotheses of Lemma 16 hold almost surely for $k = 0, \dots, N-1$. Write $\mathbb{E}_{\mathbb{Q}}$ for expectation and let $\mathcal{F}_k$ be the pre-noise history under this law. The quantities Δ_k and a_k are $\mathcal{F}_k$ -measurable because they are determined before the fresh proposal noise at step k is drawn. Suppose deterministic numbers $\underline{a}_k$ satisfy*

$$0 \leq \underline{a}_k \leq a_k \leq \frac{1}{2L}, \quad 0 \leq \mu \underline{a}_k \leq \mu a_k < 1 \quad \mathbb{Q}\text{-almost surely.}$$

If deterministic $\beta_k \geq 0$ satisfy

$$\mathbb{E}_{\mathbb{Q}}[p_k + a_k \|e_k\|_2^2 \mid \mathcal{F}_k] \leq \beta_k,$$

then

$$\mathbb{E}_{\mathbb{Q}}[\Delta_N] \leq \exp\left(-\mu \sum_{k=0}^{N-1} \underline{a}_k\right) \mathbb{E}_{\mathbb{Q}}[\Delta_0] + B_N, \quad (67)$$

where

$$B_N = \sum_{j=0}^{N-1} \exp\left(-\mu \sum_{i=j+1}^{N-1} \underline{a}_i\right) \beta_j. \quad (68)$$

Proof Equation (59) holds $\mathbb{Q}$ -almost surely. Because Δ_k is $\mathcal{F}_k$ -measurable, nonnegative, and $a_k \geq \underline{a}_k$,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[\Delta_{k+1} \mid \mathcal{F}_k] &\leq \mathbb{E}_{\mathbb{Q}}[(1 - \mu a_k)\Delta_k + p_k + a_k \|e_k\|_2^2 \mid \mathcal{F}_k] \\ &\leq (1 - \mu \underline{a}_k)\Delta_k + \beta_k. \end{aligned}$$

Taking total expectations and iterating this affine recursion gives

$$\mathbb{E}_{\mathbb{Q}}[\Delta_N] \leq \left\{ \prod_{k=0}^{N-1} (1 - \mu \underline{a}_k) \right\} \mathbb{E}_{\mathbb{Q}}[\Delta_0] + \sum_{j=0}^{N-1} \beta_j \prod_{i=j+1}^{N-1} (1 - \mu \underline{a}_i).$$

Applying $1 - z \leq e^{-z}$ to every product factor proves Equations (67)-(68). ■

For the original, unstopped stochastic process under its law $\mathbb{P}$, take $\mathbb{Q} = \mathbb{P}(\cdot \mid \Omega_\delta^{\text{loc}})$ whenever this conditional law is used. Then

$$\begin{aligned}\mathbb{E}_{\mathbb{P}}[\Delta_N] &= \mathbb{P}(\Omega_\delta^{\text{loc}})\mathbb{E}_{\mathbb{Q}}[\Delta_N] + \mathbb{E}_{\mathbb{P}}[\Delta_N \mathbf{1}_{(\Omega_\delta^{\text{loc}})^c}] \\ &\leq \mathbb{E}_{\mathbb{Q}}[\Delta_N] + \varepsilon_{\text{exit},N},\end{aligned}$$

where $\mathbf{1}_A$ denotes the indicator of an event A and $[\cdot]_+ = \max\{\cdot, 0\}$. The nonnegative exit contribution is

$$\varepsilon_{\text{exit},N} := \mathbb{E}_{\mathbb{P}}\left[[\Delta_N]_+ \mathbf{1}_{(\Omega_\delta^{\text{loc}})^c}\right]. \quad (69)$$

The positive part is needed because F_N^* is the minimum over the localized set and Δ_N need not be nonnegative after an exit. Thus an unconditional expected bound requires either a direct estimate of Equation (69) or a globally valid regularity argument. The term is retained in any unconditional expected bound.

Corollary 23 (Accuracy criterion and uniform-grid guidance budget) *Under Theorem 22, the sufficient conditions*

$$\sum_{k=0}^{N-1} a_k \geq \frac{1}{\mu} \log\left(\frac{2\mathbb{E}_{\mathbb{Q}}[\Delta_0]}{\varepsilon^*}\right), \quad B_N \leq \frac{\varepsilon^*}{2} \quad (70)$$

imply $\mathbb{E}_{\mathbb{Q}}[\Delta_N] \leq \varepsilon^*$ whenever the logarithm is positive. For the uniform grid $t_k = k/N$, Lemma 13 gives

$$\sum_{k=0}^{N-1} a_k \geq c_\zeta \rho \kappa \sum_{k=0}^{N-1} \left(1 - \frac{k}{N}\right) = c_\zeta \rho \kappa \frac{N+1}{2}. \quad (71)$$

A complexity statement beyond Equation (70) requires a separately verified rate for B_N and, under the original law, for $\varepsilon_{\text{exit},N}$.

Proof The two inequalities in Equation (70) bound the transient and defect terms in Equation (67) by $\varepsilon^*/2$. The finite sum in Equation (71) equals $(N+1)/2$. $\blacksquare$

With fixed c_ζ , increasing N changes both the number of guidance corrections and the number of fresh-noise draws; it is not ordinary ODE grid refinement. If c_ζ is scaled proportionally to the grid width, the cumulative guidance budget may remain $O(1)$, so refinement alone need not remove the optimization transient.

3.9 Implementation-specific stochastic endpoint decomposition

The preceding theorem is the shortest route to an expected bound: it treats all proposal effects through the measurable defects p_k and e_k . We now provide a second, complementary route that resolves the same stochastic step at the decoded physical endpoint for the actual re-initialization proposal. The first route is preferable when the transport defects can be measured directly; the decomposition below is useful when one wishes to attribute the error floor to endpoint bias, endpoint-Jacobian fluctuation, and curvature. It does not replace stochastic re-initialization by an additive-noise gradient model.

Throughout this subsection, expectations are taken under the localized law $\mathbb{Q}_\delta = \mathbb{P}(\cdot \mid \Omega_\delta^{\text{loc}})$, and the subscript is suppressed. Because $\Omega_\delta^{\text{loc}}$ is the intersection of independent radially symmetric truncation events, the fresh vectors remain independent of the pre-noise history, centered, and supported on $\overline{B}(0, r_\delta)$ under $\mathbb{Q}_\delta$. Define the localized second moment

$$\chi_{2,\delta} := \mathbb{E}_{\mathbb{Q}_\delta} \|\xi_k\|_2^2 \leq r_\delta^2.$$

The same value applies at each stochastic step because the truncated Gaussian vectors are identically distributed.

Let X_k denote the random model state and set

$$Z_k = E_{t_k}^\theta(X_k), \quad \psi_k = \nabla \mathcal{L}_h(Z_k), \quad J_k = J_{E_{t_k}^\theta}(X_k), \quad g_k = J_k^\top \psi_k.$$

For a stochastic proposal,

$$\begin{aligned} \overline{X}_{k+1} &= (1 - t_{k+1})\xi_k + t_{k+1}P_{t_k}^\theta(X_k), \\ X_{k+1} &= \overline{X}_{k+1} - \eta_k J_k^\top \psi_k, \quad \eta_k := \gamma_k \rho_k = a_k. \end{aligned}$$

Here ξ_k is independent of $\mathcal{F}_k$, while X_k, ψ_k, J_k , and η_k are $\mathcal{F}_k$ -measurable.

Define the unguided next endpoint and cross-time endpoint response by

$$\overline{Z}_{k+1} = E_{t_{k+1}}^\theta(\overline{X}_{k+1}), \quad B_k^E = J_{E_{t_{k+1}}^\theta}(\overline{X}_{k+1})J_k^\top.$$

Let

$$\delta_k^E = \overline{Z}_{k+1} - Z_k, \quad m_k^E = \mathbb{E}[\delta_k^E \mid \mathcal{F}_k], \quad A_k^E = \mathbb{E}[B_k^E \mid \mathcal{F}_k]. \quad (72)$$

Only the symmetric part of A_k^E contributes to $\psi_k^\top A_k^E \psi_k$.

Assumption 4 (Local endpoint differentiability for stochastic steps) *On every segment between $\overline{X}_{k+1}$ and X_{k+1} , $E_{t_{k+1}}^\theta$ is continuously differentiable and*

$$\|J_{E_{t_{k+1}}^\theta}(\mathbf{x}) - J_{E_{t_{k+1}}^\theta}(\mathbf{y})\|_{\text{op}} \leq H_E^{\text{op}} \|\mathbf{x} - \mathbf{y}\|_2.$$

Moreover, $\|J_k\|_{\text{op}} \leq \overline{J}$ on the analyzed region.

Proposition 24 (Exact decoded-endpoint decomposition) *Under Assumption 4, assume $\overline{Z}_{k+1}$ and B_k^E are conditionally integrable. Then there is a Taylor remainder $\mathbf{t}_k^E$ such that*

$$Z_{k+1} = \overline{Z}_{k+1} - \eta_k B_k^E \psi_k + \mathbf{t}_k^E, \quad (73)$$

$$\|\mathbf{t}_k^E\|_2 \leq \bar{\mathbf{t}}_k := \frac{H_E^{\text{op}}}{2} \eta_k^2 \overline{J}^2 \|\psi_k\|_2^2. \quad (74)$$

Consequently,

$$Z_{k+1} = Z_k - \eta_k A_k^E \psi_k + b_k^E + \varepsilon_k^E, \quad (75)$$

where

$$b_k^E = m_k^E + \mathbb{E}[\mathbf{t}_k^E \mid \mathcal{F}_k], \quad (76)$$

$$\varepsilon_k^E = \{\delta_k^E - m_k^E\} - \eta_k \{B_k^E - A_k^E\} \psi_k + \{\mathbf{t}_k^E - \mathbb{E}[\mathbf{t}_k^E \mid \mathcal{F}_k]\}, \quad (77)$$

and

$$\mathbb{E}[\varepsilon_k^E \mid \mathcal{F}_k] = 0. \quad (78)$$

Proof Put $d_k^{\text{corr}} = -\eta_k J_k^\top \psi_k$. The integral Taylor formula gives

$$\begin{aligned} E_{t_{k+1}}^\theta(\bar{X}_{k+1} + d_k^{\text{corr}}) &= E_{t_{k+1}}^\theta(\bar{X}_{k+1}) + J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1})d_k^{\text{corr}} + \mathfrak{t}_k^E, \\ \mathfrak{t}_k^E &= \int_0^1 \{J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1} + sd_k^{\text{corr}}) - J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1})\}d_k^{\text{corr}} ds. \end{aligned}$$

Assumption 4 yields $\|\mathfrak{t}_k^E\|_2 \leq H_E^{\text{op}} \|d_k^{\text{corr}}\|_2^2/2$, which implies Equation (74). The linear term is $-\eta_k B_k^E \psi_k$, proving Equation (73). Substitute $\bar{Z}_{k+1} = Z_k + \delta_k^E$, add and subtract the conditional means in Equation (72), and collect the centered terms. Each term in Equation (77) has zero conditional mean. $\blacksquare$

The three centered terms arise from different mechanisms: fresh endpoint fluctuation, endpoint-Jacobian fluctuation, and endpoint curvature under the guidance displacement.

Lemma 25 (Explicit localized bias and variance bounds) *Let*

$$\delta_{k+1}^{\text{noise}} = 1 - t_{k+1}, \quad \mathbf{x}_{k+1}^{\text{ctr}} = t_{k+1} P_{t_k}^\theta(X_k), \quad \kappa_k = \|E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}}) - Z_k\|_2. \quad (79)$$

In addition to Assumption 4, suppose $E_{t_{k+1}}^\theta$ is M_E -Lipschitz and its Jacobian is H_E^{op} -Lipschitz in operator norm and H_E^{F} -Lipschitz in Frobenius norm on

$$\mathbf{x}_{k+1}^{\text{ctr}} + \delta_{k+1}^{\text{noise}} \bar{B}(0, r_\delta)$$

and on the subsequent guidance segments. By Corollary 8, assume the localized endpoint region is chosen so that $\|\psi_k\|_2 \leq G_{\mathcal{L}}$. Set

$$\nu_{E,k}^2 = \mathbb{E}[\|\delta_k^E - m_k^E\|_2^2 \mid \mathcal{F}_k], \quad (80)$$

$$\nu_{J,k}^2 = \mathbb{E}[\|B_k^E - A_k^E\|_{\text{F}}^2 \mid \mathcal{F}_k]. \quad (81)$$

Here $\|\cdot\|_{\text{F}}$ denotes the Frobenius norm. Then

$$\|m_k^E\|_2 \leq \kappa_k + \frac{H_E^{\text{op}}}{2} (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta}, \quad (82)$$

$$\nu_{E,k}^2 \leq M_E^2 (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta}, \quad (83)$$

$$\nu_{J,k}^2 \leq (H_E^{\text{F}})^2 (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta} \bar{J}^2. \quad (84)$$

Define

$$\bar{\mathfrak{t}}_k^{\text{loc}} = \frac{H_E^{\text{op}}}{2} \eta_k^2 \bar{J}^2 G_{\mathcal{L}}^2. \quad (85)$$

Then the bias and centered fluctuation in Proposition 24 satisfy

$$\|b_k^E\|_2 \leq \bar{b}_k^E := \kappa_k + \frac{H_E^{\text{op}}}{2} (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta} + \bar{\mathfrak{t}}_k^{\text{loc}}, \quad (86)$$

$$\mathbb{E}[\|\varepsilon_k^E\|_2^2 \mid \mathcal{F}_k] \leq v_k^2 := 3\nu_{E,k}^2 + 3\eta_k^2 \nu_{J,k}^2 G_{\mathcal{L}}^2 + 3(\bar{\mathfrak{t}}_k^{\text{loc}})^2. \quad (87)$$

Proof Conditionally on $\mathcal{F}_k$ under the localized law,

$$\bar{X}_{k+1} = \mathbf{x}_{k+1}^{\text{ctr}} + \delta_{k+1}^{\text{noise}} \xi_k, \quad \mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0, \quad \mathbb{E}[\|\xi_k\|_2^2 \mid \mathcal{F}_k] = \chi_{2,\delta}.$$

Taylor's formula at $\mathbf{x}_{k+1}^{\text{ctr}}$ gives

$$\begin{aligned} E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}} + \delta_{k+1}^{\text{noise}} \xi_k) \\ = E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}}) + \delta_{k+1}^{\text{noise}} J_{E_{t_{k+1}}^\theta}(\mathbf{x}_{k+1}^{\text{ctr}}) \xi_k + R_k^{\text{noise}}, \end{aligned}$$

where

$$\|R_k^{\text{noise}}\|_2 \leq \frac{H_E^{\text{op}}}{2} (\delta_{k+1}^{\text{noise}})^2 \|\xi_k\|_2^2.$$

Taking conditional expectation eliminates the linear term. Adding and subtracting $E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}})$ and using the definition of κ_k therefore yields

$$\|m_k^E\|_2 \leq \kappa_k + \frac{H_E^{\text{op}}}{2} (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta},$$

which proves Equation (82).

For the endpoint fluctuation, conditional variance is no larger than the mean-square distance to any $\mathcal{F}_k$ -measurable reference. Choosing $E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}})$ gives

$$\begin{aligned} \nu_{E,k}^2 &\leq \mathbb{E} \left[\left\| E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}} + \delta_{k+1}^{\text{noise}} \xi_k) - E_{t_{k+1}}^\theta(\mathbf{x}_{k+1}^{\text{ctr}}) \right\|_2^2 \mid d\ell e \mid \mathcal{F}_k \right] \\ &\leq M_E^2 (\delta_{k+1}^{\text{noise}})^2 \mathbb{E}[\|\xi_k\|_2^2 \mid \mathcal{F}_k] \leq M_E^2 (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta}. \end{aligned}$$

Similarly, conditional variance in the Frobenius Hilbert space is bounded by the mean-square distance to $J_{E_{t_{k+1}}^\theta}(\mathbf{x}_{k+1}^{\text{ctr}}) J_k^\top$. The Jacobian Lipschitz condition and $\|J_k\|_{\text{op}} \leq \bar{J}$ give

$$\begin{aligned} \nu_{J,k}^2 &\leq (H_E^{\text{F}})^2 (\delta_{k+1}^{\text{noise}})^2 \bar{J}^2 \mathbb{E}[\|\xi_k\|_2^2 \mid \mathcal{F}_k] \\ &\leq (H_E^{\text{F}})^2 (\delta_{k+1}^{\text{noise}})^2 \chi_{2,\delta} \bar{J}^2, \end{aligned}$$

proving Equation (84).

Equation (74) and the localized bounds on ψ_k and J_k imply $\|\mathbf{t}_k^E\|_2 \leq \bar{\mathbf{t}}_k^{\text{loc}}$. The triangle inequality then proves Equation (86). Finally,

$$\|a + b + c\|_2^2 \leq 3(\|a\|_2^2 + \|b\|_2^2 + \|c\|_2^2),$$

$\|(B - A)\psi\|_2 \leq \|B - A\|_{\text{F}} \|\psi\|_2$, and

$$\mathbb{E}[\|\mathbf{t}_k^E - \mathbb{E}[\mathbf{t}_k^E \mid \mathcal{F}_k]\|_2^2 \mid \mathcal{F}_k] \leq \mathbb{E}[\|\mathbf{t}_k^E\|_2^2 \mid \mathcal{F}_k] \leq (\bar{\mathbf{t}}_k^{\text{loc}})^2$$

produce Equation (87) from Equation (77). ■

The derivative assumptions above are required only on the localized bounded noise slice and the associated guidance segments. Under the original Gaussian law, an analogous statement would require global derivative control or an additional tail remainder.

Assumption 5 (Direction-specific effective endpoint descent) *Let $\mathcal{Z}_\delta^{\text{stoch}}$ be a convex endpoint region that contains every segment $[Z_k, Z_{k+1}]$. Define*

$$\mathcal{L}_h^\star = \inf_{z \in \mathcal{Z}_\delta^{\text{stoch}}} \mathcal{L}_h(z). \quad (88)$$

On $\mathcal{Z}_\delta^{\text{stoch}}$, $\mathcal{L}_h$ has an $L_{\mathcal{L}}$ -Lipschitz gradient. Assume that, for some $\lambda_A > 0$,

$$\psi_k^\top A_k^E \psi_k \geq \lambda_A \|\psi_k\|_2^2 \quad (89)$$

almost surely on every analyzed step. If $\|J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1})\|_{\text{op}} \leq M_E$ and $\|J_k\|_{\text{op}} \leq \bar{J}$ on the localized region, then conditional Jensen gives the automatic upper bound

$$\|A_k^E\|_{\text{op}} \leq M_E \bar{J} =: M_A. \quad (90)$$

Indeed,

$$\begin{aligned} \|A_k^E\|_{\text{op}} &= \|\mathbb{E}[B_k^E \mid \mathcal{F}_k]\|_{\text{op}} \\ &\leq \mathbb{E} \left[\|J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1})\|_{\text{op}} \|J_k\|_{\text{op}} \mid \mathcal{F}_k \right] \leq M_E \bar{J}. \end{aligned}$$

Equation (89) asks only for positive conditional mean response in the *realized physical guidance direction*. It is weaker than a uniform lower eigenvalue bound for all vectors, states, and flow-time pairs, but it is still a conditional property of the trained model.

Lemma 26 (A direction-specific Jacobian diagnostic) *Let $G_k = J_k J_k^\top$. Suppose*

$$\psi_k^\top G_k \psi_k \geq \lambda_{J,\psi} \|\psi_k\|_2^2, \quad \|J_k\|_{\text{op}} \leq \bar{J}, \quad (91)$$

and

$$\mathbb{E} \left[\|J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1}) - J_k\|_{\text{op}} \mid \mathcal{F}_k \right] \leq \epsilon_{J,k}. \quad (92)$$

Then

$$\psi_k^\top A_k^E \psi_k \geq (\lambda_{J,\psi} - \epsilon_{J,k} \bar{J}) \|\psi_k\|_2^2. \quad (93)$$

Hence Assumption 5 holds on that step if the right-hand coefficient is positive.

Proof By definition,

$$A_k^E - G_k = \mathbb{E}[\{J_{E_{t_{k+1}}^\theta}(\bar{X}_{k+1}) - J_k\} J_k^\top \mid \mathcal{F}_k],$$

so conditional Jensen and submultiplicativity give $\|A_k^E - G_k\|_{\text{op}} \leq \epsilon_{J,k} \bar{J}$. Therefore

$$\psi_k^\top A_k^E \psi_k \geq \psi_k^\top G_k \psi_k - \|A_k^E - G_k\|_{\text{op}} \|\psi_k\|_2^2,$$

which proves the result. ■

Theorem 27 (One-step physical-endpoint descent) *Suppose Proposition 24 and Assumption 5 hold. Assume*

$$\|b_k^E\|_2 \leq \bar{b}_k^E, \quad \mathbb{E}[\|\varepsilon_k^E\|_2^2 \mid \mathcal{F}_k] \leq v_k^2, \quad (94)$$

where the bounds are $\mathcal{F}_k$ -measurable. If $\eta_k > 0$ and

$$\eta_k \leq \frac{\lambda_A}{6L_{\mathcal{L}}M_A^2}, \quad (95)$$

with the restriction vacuous when $L_{\mathcal{L}} = 0$, then

$$\mathbb{E}[\mathcal{L}_h(Z_{k+1}) \mid \mathcal{F}_k] \leq \mathcal{L}_h(Z_k) - \frac{\eta_k \lambda_A}{2} \|\psi_k\|_2^2 + \mathfrak{e}_k, \quad (96)$$

where

$$\mathfrak{e}_k = \frac{(\bar{b}_k^E)^2}{\eta_k \lambda_A} + \frac{3L_{\mathcal{L}}}{2} ((\bar{b}_k^E)^2 + v_k^2). \quad (97)$$

The explicit quantities in Lemma 25 may be substituted into this error.

Proof By Equation (75),

$$D_k := Z_{k+1} - Z_k = -\eta_k A_k^E \psi_k + b_k^E + \varepsilon_k^E.$$

The descent lemma for $\mathcal{L}_h$ gives

$$\mathcal{L}_h(Z_{k+1}) \leq \mathcal{L}_h(Z_k) + \psi_k^\top D_k + \frac{L_{\mathcal{L}}}{2} \|D_k\|_2^2.$$

After conditional expectation, the centered error vanishes. Effective alignment and Young's inequality give

$$-\eta_k \psi_k^\top A_k^E \psi_k + \psi_k^\top b_k^E \leq -\frac{3\eta_k \lambda_A}{4} \|\psi_k\|_2^2 + \frac{(\bar{b}_k^E)^2}{\eta_k \lambda_A}.$$

Moreover,

$$\mathbb{E}[\|D_k\|_2^2 \mid \mathcal{F}_k] \leq 3\eta_k^2 M_A^2 \|\psi_k\|_2^2 + 3(\bar{b}_k^E)^2 + 3v_k^2.$$

The step-size restriction bounds the first quadratic term by $\eta_k \lambda_A \|\psi_k\|_2^2 / 4$. Combining the estimates proves Equations (96)-(97). $\blacksquare$

Corollary 28 (Finite-step physical stationarity) *Assume Theorem 27 and $\mathbb{E}[\mathcal{L}_h(Z_0) - \mathcal{L}_h^*] < \infty$. Then, for any finite N for which $\sum_{k=0}^{N-1} \mathbb{E}[\mathfrak{e}_k] < \infty$,*

$$\frac{\lambda_A}{2} \sum_{k=0}^{N-1} \mathbb{E}[\eta_k \|\psi_k\|_2^2] \leq \mathbb{E}[\mathcal{L}_h(Z_0) - \mathcal{L}_h^*] + \sum_{k=0}^{N-1} \mathbb{E}[\mathfrak{e}_k]. \quad (98)$$

If $\eta_k \geq \underline{\eta}_k > 0$ almost surely and $\Gamma_N^S = \sum_{k=0}^{N-1} \underline{\eta}_k$, then

$$\min_{0 \leq k < N} \mathbb{E}[\|\psi_k\|_2^2] \leq \frac{2 \left(\mathbb{E}[\mathcal{L}_h(Z_0) - \mathcal{L}_h^*] + \sum_{k=0}^{N-1} \mathbb{E}[\mathfrak{e}_k] \right)}{\lambda_A \Gamma_N^S}. \quad (99)$$

For local assumptions, the same estimate holds with indicators before a stopping time at first exit; an unstopped claim requires additional tail control.

Proof Take total expectations in Equation (96) and sum. The physical loss telescopes and is bounded below by $\mathcal{L}_h^*$. The minimum-gradient bound follows from $\eta_k \geq \underline{\eta}_k$. $\blacksquare$

Assumption 6 (Local PL geometry in physical endpoint space) *On the same region $\mathcal{Z}_\delta^{\text{stoch}}$ used in Assumption 5,*

$$\|\nabla \mathcal{L}_h(\mathbf{z})\|_2^2 \geq 2\mu_{\mathbf{z}}\{\mathcal{L}_h(\mathbf{z}) - \mathcal{L}_h^*\} \quad (100)$$

for some $\mu_{\mathbf{z}} > 0$.

Corollary 29 (Stochastic contraction above an explicit endpoint floor) *Under Theorem 27 and Assumption 6, suppose $\eta_k \geq \underline{\eta}_k > 0$ and set*

$$\chi_k^S = \mu_{\mathbf{z}} \lambda_A \underline{\eta}_k \in (0, 1), \quad \bar{\epsilon}_k = \mathbb{E}[\epsilon_k]. \quad (101)$$

For $V_k = \mathbb{E}[\mathcal{L}_h(Z_k) - \mathcal{L}_h^*]$,

$$V_{k+1} \leq (1 - \chi_k^S)V_k + \bar{\epsilon}_k, \quad (102)$$

and therefore

$$V_N \leq V_0 \prod_{k=0}^{N-1} (1 - \chi_k^S) + \sum_{j=0}^{N-1} \bar{\epsilon}_j \prod_{k=j+1}^{N-1} (1 - \chi_k^S). \quad (103)$$

If $\chi_k^S \geq \underline{\chi}^S \in (0, 1)$ and $\bar{\epsilon}_k \leq \bar{\epsilon}$, then

$$V_N \leq (1 - \underline{\chi}^S)^N V_0 + \underbrace{\frac{\bar{\epsilon}}{\underline{\chi}^S}}_{=: \epsilon_S^{\text{floor}}}. \quad (104)$$

A target is certified only when it is strictly larger than $\epsilon_S^{\text{floor}}$.

Proof Apply Equation (100) to the descent term in Equation (96), use $\eta_k \geq \underline{\eta}_k$, and take total expectations. Iteration gives the product bound; the uniform result follows from the geometric series. $\blacksquare$

Substituting Lemma 25 into Equation (97) shows explicitly that the stochastic floor contains: deterministic endpoint-consistency error κ_k , Gaussian smoothing bias, endpoint fluctuation, endpoint-Jacobian fluctuation, and second-order endpoint curvature. Shrinking η_k alone does not remove a bias $\bar{b}_k^E$ unless that bias shrinks at a compatible rate because $(\bar{b}_k^E)^2/(\eta_k \lambda_A)$ appears in the bound.

3.10 Hybrid composition

Theorem 30 (Two-phase hybrid bound) *Suppose two consecutive phases, under the same probability or localization law and for the same nonnegative gap Δ_k , satisfy*

$$\begin{aligned} \mathbb{E}\Delta_K &\leq Q_1 \mathbb{E}\Delta_0 + R_1, \\ \mathbb{E}\Delta_N &\leq Q_2 \mathbb{E}\Delta_K + R_2, \quad Q_1, Q_2 \in [0, 1]. \end{aligned} \quad (105)$$

Then

$$\mathbb{E}\Delta_N \leq Q_2 Q_1 \mathbb{E}\Delta_0 + Q_2 R_1 + R_2. \quad (106)$$

The formula applies to either phase ordering. It does not imply that a hybrid uniformly improves on a pure sampler. A phase is beneficial only when its additional contraction outweighs the defect that it introduces and passes to the next phase.

Proof Starting from the second phase bound and applying the first phase bound,

$$\begin{aligned} \mathbb{E}\Delta_N &\leq Q_2 \mathbb{E}\Delta_K + R_2 \\ &\leq Q_2 \{Q_1 \mathbb{E}\Delta_0 + R_1\} + R_2 \\ &= Q_2 Q_1 \mathbb{E}\Delta_0 + Q_2 R_1 + R_2. \end{aligned}$$

The derivation uses only the two phase-wise inequalities and is therefore unchanged when the order of the deterministic and stochastic phases is reversed, provided Q_i and R_i are assigned to the corresponding phases. $\blacksquare$

Equation (106) provides a composition rule rather than an ordering of sampler phases. Whether a deterministic warm start improves a pure stochastic run depends on the phase-specific contraction and defect terms.

3.11 From the discrete guidance loss to a physical solution set

The preceding bounds control the finite-dimensional loss evaluated by the sampler. A separate consistency and identifiability step is needed to translate such a loss bound into distance from a reference physical solution set. Let $\mathcal{L}_{\text{ref}}$ be an exact or reference physical objective on

$$\mathcal{Z}_{\mathcal{K}}^{\text{term}} = E_1^\theta(\mathcal{K}) = \mathcal{D}_{\text{phys}}(\mathcal{K}), \quad \mathcal{S}_{\mathcal{K}} = \arg \min_{z \in \mathcal{Z}_{\mathcal{K}}^{\text{term}}} \mathcal{L}_{\text{ref}}(z),$$

Because $\mathcal{K}$ is compact and $\mathcal{D}_{\text{phys}}$ is affine, $\mathcal{Z}_{\mathcal{K}}^{\text{term}}$ is compact. The local smoothness conditions imply that $\mathcal{L}_h$ is continuous on this set and therefore attains its minimum there. Assume in addition that $\mathcal{L}_{\text{ref}}$ is continuous on the compact set $\mathcal{Z}_{\mathcal{K}}^{\text{term}}$, so $\mathcal{S}_{\mathcal{K}}$ is nonempty, and define

$$\mathcal{L}_{\text{ref}, \mathcal{K}}^* = \min_{z \in \mathcal{Z}_{\mathcal{K}}^{\text{term}}} \mathcal{L}_{\text{ref}}(z).$$

The set $\mathcal{S}_{\mathcal{K}}$ may contain multiple coefficient-solution pairs; uniqueness is not assumed.

Assumption 7 (Discrete consistency and local set identifiability) *There are constants $\varepsilon_{\text{phys}} \geq 0$ and $C_{\text{id}} > 0$ such that, for all $z \in \mathcal{Z}_{\mathcal{K}}^{\text{term}}$,*

$$\begin{aligned} |\mathcal{L}_h(z) - \mathcal{L}_{\text{ref}}(z)| &\leq \varepsilon_{\text{phys}}, \\ \text{dist}(z, \mathcal{S}_{\mathcal{K}})^2 &\leq C_{\text{id}} \{\mathcal{L}_{\text{ref}}(z) - \mathcal{L}_{\text{ref}, \mathcal{K}}^*\}. \end{aligned} \quad (107)$$

The consistency term may contain spatial-discretization error, endpoint-surrogate error, boundary- and initial-condition approximation, observation noise relative to the chosen reference objective, and mismatch between the data-generating solver and the residual operator.

Theorem 31 (Conditional distance to the physical solution set) *Suppose $\mathbf{x}_N \in \mathcal{K}$ under the probability law of the applicable algorithmic bound and Assumption 7 holds. Let $\mathbf{z}_N = \mathcal{D}_{\text{phys}}(\mathbf{x}_N)$ and let $\Delta_N = F_N(\mathbf{x}_N) - F_N^*$ be defined by Equation (33). Then*

$$\mathbb{E}[\text{dist}(\mathbf{z}_N, \mathcal{S}_{\mathcal{K}})^2] \leq C_{\text{id}} \{\mathbb{E}[\Delta_N] + 2\varepsilon_{\text{phys}}\}.$$

Proof Set

$$\mathcal{L}_{h,\mathcal{K}}^* = \min_{\mathbf{z} \in \mathcal{Z}_{\mathcal{K}}^{\text{term}}} \mathcal{L}_h(\mathbf{z}).$$

Because P_1^θ is the identity and $F_N(\mathbf{x}) = \mathcal{L}_h(\mathcal{D}_{\text{phys}}(\mathbf{x}))$, minimizing F_N over $\mathcal{K}$ is equivalent to minimizing $\mathcal{L}_h$ over $\mathcal{Z}_{\mathcal{K}}^{\text{term}}$; hence $F_N^* = \mathcal{L}_{h,\mathcal{K}}^*$. Uniform consistency gives

$$\mathcal{L}_{\text{ref}}(\mathbf{z}_N) \leq \mathcal{L}_h(\mathbf{z}_N) + \varepsilon_{\text{phys}}.$$

If $\mathbf{z}_h^*$ minimizes $\mathcal{L}_h$ over $\mathcal{Z}_{\mathcal{K}}^{\text{term}}$, then

$$\mathcal{L}_{\text{ref},\mathcal{K}}^* \leq \mathcal{L}_{\text{ref}}(\mathbf{z}_h^*) \leq \mathcal{L}_h(\mathbf{z}_h^*) + \varepsilon_{\text{phys}} = \mathcal{L}_{h,\mathcal{K}}^* + \varepsilon_{\text{phys}}.$$

Equivalently, $-\mathcal{L}_{\text{ref},\mathcal{K}}^* \geq -\mathcal{L}_{h,\mathcal{K}}^* - \varepsilon_{\text{phys}}$. Combining the two inequalities yields

$$\begin{aligned} \mathcal{L}_{\text{ref}}(\mathbf{z}_N) - \mathcal{L}_{\text{ref},\mathcal{K}}^* &\leq \mathcal{L}_h(\mathbf{z}_N) - \mathcal{L}_{h,\mathcal{K}}^* + 2\varepsilon_{\text{phys}} \\ &= \Delta_N + 2\varepsilon_{\text{phys}}. \end{aligned}$$

Apply Equation (107) pointwise and take expectations under the same law as the algorithmic bound. ■

For a non-identifiable inverse problem, the appropriate conclusion is necessarily set-valued and Equation (107) may fail. For endpoint-only temporal residuals, $\varepsilon_{\text{phys}}$ includes the Hermite surrogate contribution characterized in Proposition 1. For the scalar-vorticity Navier-Stokes problem, the reference semidiscrete operator is the corrected vorticity-transport right-hand side described in Section 4.1; endpoint-only temporal approximation remains part of $\varepsilon_{\text{phys}}$.

4 Experiments

We evaluate FM4PDE against a broad set of external baselines on forward, inverse, sparse-reconstruction, sparse-inverse, and time-dependent data-assimilation settings. Unless stated otherwise, the main experimental matrix uses the 11 PDE families: Darcy Flow, Poisson Equations, Helmholtz Equations, Navier-Stokes Equations, Burgers Equations, Reaction-Diffusion Equations, Shallow-Water Equations, Heat Equations, Wave Equations, Advection-Diffusion Equations, and Steady Heat Conduction Equations.

4.1 Data Preparation

Let $\tau \in [0, \mathcal{T}]$ denote the physical time variable and $\mathbf{c} \in \Omega$ represent the spatial coordinates, with $\partial\Omega$ defining the domain boundary. Each PDE dataset comprises $N_{\text{train}} = 50,000$ training samples and $N_{\text{test}} = 10,000$ testing samples. One-dimensional equations are discretized

with a spatial and temporal resolution of 128, whereas two-dimensional equations utilize a spatial resolution of 128×128 across 10 temporal snapshots. Training and testing sets are generated using mutually exclusive random seeds to ensure rigorous evaluation.

For a predicted physical state $\mathbf{z} = (\mathbf{a}, \mathbf{u})$ in a static problem, the PDE residual loss is computed from an interior residual r_Ω and the corresponding boundary residual $r_{\partial\Omega}$. For time-dependent endpoint datasets with initial and terminal states q_0 and $q_\mathcal{T}$ at the physical horizon $\mathcal{T}$, we express the governing equation in first-order form:

$$\partial_\tau q = F(q; \theta),$$

where θ denotes scalar physical parameters such as diffusivity, wave speed, or advection velocity. In the endpoint-only setting, the primary residual is evaluated using an endpoint-induced cubic Hermite bridge. Let $F_0 = F(q_0; \theta)$, $F_\mathcal{T} = F(q_\mathcal{T}; \theta)$, and the interpolation parameter $s \in [0, 1]$. The bridge is formulated as:

$$H(s) = (2s^3 - 3s^2 + 1)q_0 + (s^3 - 2s^2 + s)\mathcal{T}F_0 + (-2s^3 + 3s^2)q_\mathcal{T} + (s^3 - s^2)\mathcal{T}F_\mathcal{T}.$$

At collocation points $s_j \in (0, 1)$, the interior residual is given by:

$$r_{\Omega,j} = \mathcal{T}^{-1} \frac{\partial H}{\partial s}(s_j) - F(H(s_j); \theta).$$

Boundary residuals are evaluated consistently with the data-generating boundary conditions at the endpoints and, when applicable, at the bridge collocation states. Periodic boundary conditions are enforced through wrap-around consistency, incorporating derivative-continuity constraints. Dirichlet and Neumann conditions are enforced via direct boundary-value matches and one-sided finite-difference residuals, respectively. When an observed initial condition is provided during a forward or joint task, an initial-condition residual is strictly applied over the observed initial-state mask. The total PDE loss is aggregated component-wise as:

$$L_{\text{pde}} = \text{MSE}(r_\Omega) + \lambda_{\partial\Omega} \text{MSE}(r_{\partial\Omega}) + \lambda_{\text{ic}} \text{MSE}(r_{\text{ic}}) + \lambda_{\text{ep}} \text{MSE}(r_{\text{ep}}),$$

where terms inapplicable to a specific PDE are omitted. All observation losses and PDE residuals are evaluated following inverse standardization—that is, strictly within the original physical units rather than normalized network coordinates.

Darcy flow. Let $(\mathbf{a}, \mathbf{u}) \triangleq (a, p)$. We consider the elliptic Darcy problem on $\Omega = [0, 1]^2$:

$$\begin{aligned} -\nabla \cdot (a(\mathbf{c}) \nabla p(\mathbf{c})) &= 1, \quad \mathbf{c} = (x, y) \in \Omega, \\ p(\mathbf{c}) &= 0, \quad \mathbf{c} \in \partial\Omega. \end{aligned}$$

The coefficient field is obtained by thresholding a Gaussian random field (GRF): $a(\mathbf{c}) = 12$ where the latent GRF is strictly nonnegative, and $a(\mathbf{c}) = 4$ otherwise. The data generation process employs GRF hyperparameters $\alpha = 3$ and $\tau = 4$, solving the elliptic equation via a finite-volume and finite-difference groundwater flow solver. The governing residual is:

$$r_\Omega = \nabla \cdot (a \nabla p) + 1, \quad r_{\partial\Omega} = p|_{\partial\Omega}.$$

Poisson equation. Let $(\mathbf{a}, \mathbf{u}) \triangleq (f, \phi)$. We consider the Poisson problem on $\Omega = [0, 1]^2$:

$$\begin{aligned}\Delta\phi(\mathbf{c}) &= f(\mathbf{c}), \quad \mathbf{c} \in \Omega, \\ \phi(\mathbf{c}) &= 0, \quad \mathbf{c} \in \partial\Omega.\end{aligned}$$

The source field f is sampled from a GRF with $\alpha = 3$ and $\tau = 4$. The reference solution is computed using a five-point finite-difference linear solver. The residual is defined as:

$$r_\Omega = \Delta\phi - f, \quad r_{\partial\Omega} = \phi|_{\partial\Omega}.$$

Helmholtz equation. Let $(\mathbf{a}, \mathbf{u}) \triangleq (f, \psi)$. The inhomogeneous Helmholtz problem on $\Omega = [0, 1]^2$ is given by:

$$\begin{aligned}\Delta\psi(\mathbf{c}) + k^2\psi(\mathbf{c}) &= f(\mathbf{c}), \quad \mathbf{c} \in \Omega, \\ \psi(\mathbf{c}) &= 0, \quad \mathbf{c} \in \partial\Omega,\end{aligned}$$

with the wave number fixed at $k = 1$ across all experiments. The source term f is drawn from the identical GRF family utilized for the Poisson dataset. The solution is obtained by solving a Kronecker-product two-dimensional Laplacian linear system with homogeneous Dirichlet constraints. The corresponding residual is:

$$r_\Omega = \Delta\psi + k^2\psi - f, \quad r_{\partial\Omega} = \psi|_{\partial\Omega}.$$

Periodic Navier-Stokes in vorticity form. Let $(\mathbf{a}, \mathbf{u}) \triangleq (\omega_0, \omega_\mathcal{T})$. The Navier-Stokes dataset is generated on a periodic torus $\Omega = [0, 1]^2$ using the scalar-vorticity formulation:

$$\begin{aligned}\partial_\tau\omega + v_x\partial_x\omega + v_y\partial_y\omega &= \nu\Delta\omega + f_{\text{NS}}(x, y), \\ -\Delta\psi &= \omega, \quad \mathbf{v} = (v_x, v_y) = (\partial_y\psi, -\partial_x\psi).\end{aligned}$$

The external forcing field is analytically prescribed as:

$$f_{\text{NS}}(x, y) = 0.1\{\sin(2\pi(x + y)) + \cos(2\pi(x + y))\},$$

with kinematic viscosity $\nu = 10^{-3}$, a final integration time of $\mathcal{T} = 1$, an internal solver step of $\Delta\tau = 10^{-4}$, and 10 recorded temporal intervals. The initial vorticity is sampled from a two-dimensional GRF. The training partition utilizes GRF parameters $(\alpha, \tau) = (2.5, 7.0)$, while the testing partition is drawn from $(\alpha, \tau) = (3.0, 6.5)$. The physical vorticity-transport residual is formulated as:

$$r_\Omega = \partial_\tau\omega + \mathbf{v} \cdot \nabla\omega - \nu\Delta\omega - f_{\text{NS}}.$$

Burgers' equation. Let $\mathbf{x} = (u_0, u_1, \dots, u_\mathcal{T})$. The Burgers dataset captures the dynamics of the one-dimensional viscous Burgers' equation on $\Omega = [0, 1]$:

$$\partial_\tau u + \frac{1}{2}\partial_x(u^2) = \nu\partial_{xx}u, \quad x \in \Omega, \quad \nu = 10^{-2},$$

subject to periodic spatial boundary conditions. Initial conditions are sampled from a periodic GRF with $\gamma = 2.5$, $\tau = 7$, and $\sigma = 7^2$. The numerical solver extracts 128 temporal nodes over $[0, 1]$ and 128 spatial grid points, formulating the resulting field as a holistic 128×128 spatio-temporal tensor. Because the complete temporal trajectory is retained, the residual is constructed using finite differences across the spatio-temporal domain rather than relying on the endpoint Hermite bridge:

$$r_\Omega = \partial_\tau u + u\partial_x u - 10^{-2}\partial_{xx}u,$$

with strict periodic residuals enforced along the spatial axis.

Reaction-diffusion system. Let $(\mathbf{a}, \mathbf{u}) \triangleq (u_0, v_0, u_{\mathcal{T}}, v_{\mathcal{T}})$. The reaction-diffusion dataset simulates a two-component FitzHugh-Nagumo-type system on $\Omega = [-1, 1]^2$:

$$\begin{aligned}\partial_{\tau}u &= D_u\Delta u + u - u^3 - k - v, \\ \partial_{\tau}v &= D_v\Delta v + u - v,\end{aligned}$$

enforcing homogeneous Neumann boundary conditions $\nabla u \cdot \mathbf{n} = \nabla v \cdot \mathbf{n} = 0$ on $\partial\Omega$. The baseline physical parameters are defined as:

$$D_u = 2 \times 10^{-3}, \quad D_v = 4 \times 10^{-3}, \quad k = 3 \times 10^{-3}, \quad \mathcal{T} = 1.$$

The numerical integration spans 11 temporal frames to yield 10 discrete intervals. Initial states are sampled from smooth GRFs. The endpoint Hermite residual applies $q = (u, v)$ mapping to:

$$F(q) = \begin{pmatrix} D_u\Delta u + u - u^3 - k - v \\ D_v\Delta v + u - v \end{pmatrix},$$

complemented by Neumann boundary residuals for both species u and v .

Shallow-water equations. Let $(\mathbf{a}, \mathbf{u}) \triangleq (h_0, m_{x,0}, m_{y,0}, h_{\mathcal{T}}, m_{x,T}, m_{y,T})$. The shallow-water dataset captures fluid dynamics evolving from a radial dam-break initialization, utilizing conservative variables $q = (h, m_x, m_y) = (h, hu, hv)$, where h denotes water depth and (m_x, m_y) represent the momentum components. The governing balance laws are:

$$\partial_{\tau}q + \partial_x F_x(q) + \partial_y F_y(q) = 0,$$

incorporating a gravitational constant $g = 1$ and fluxes:

$$F_x(q) = \begin{pmatrix} m_x \\ m_x^2/h + \frac{1}{2}gh^2 \\ m_x m_y/h \end{pmatrix}, \quad F_y(q) = \begin{pmatrix} m_y \\ m_x m_y/h \\ m_y^2/h + \frac{1}{2}gh^2 \end{pmatrix}.$$

To guarantee numerical stability, the depth h in the denominator is regularized as $h_{\varepsilon} = \max(h, \varepsilon_h)$ using a minor constant $\varepsilon_h > 0$. Each simulation initiates with a radial dam of radius $R \sim \mathcal{U}(0.4, 0.8)$ and an interior fluid height $h_{\text{in}} \sim \mathcal{U}(2, 3)$, surrounded by a resting exterior depth of 1 and zero initial momentum. The simulation spans a physical horizon of $\mathcal{T} = 1$ divided across 10 intervals, strictly enforcing an open (zero-order Neumann extrapolation) boundary condition. The endpoint Hermite residual evaluates:

$$F(q) = -\partial_x F_x(q) - \partial_y F_y(q),$$

while the boundary residuals directly align with the open-boundary extrapolations applied by the solver. The physical positivity of depth h is tracked independently and is not supplanted by nonphysical momentum residuals.

Heat equation. Let $(\mathbf{a}, \mathbf{u}) \triangleq (u_0, u_{\mathcal{T}})$. The heat equation dataset models the thermal diffusion process governed by:

$$\partial_{\tau}u = \alpha\Delta u, \quad (x, y) \in [0, 1]^2,$$

subject to periodic boundary conditions. The initial thermal field u_0 is drawn from a smooth periodic GRF, and the thermal diffusivity is uniformly sampled as $\alpha \sim \mathcal{U}(5 \times 10^{-4}, 5 \times 10^{-3})$. The exact analytical solution is computed spectrally via Fourier transforms at 11 temporal nodes up to a final time of $\mathcal{T} = 1$. The diffusivity α is intrinsically preserved as a scalar physical parameter characterizing the system. The corresponding endpoint Hermite residual evaluates:

$$F(u) = \alpha \Delta u,$$

ensuring strict periodic boundary consistency.

Wave equation. Let $(\mathbf{a}, \mathbf{u}) \triangleq (u_0, v_0, u_{\mathcal{T}}, v_{\mathcal{T}})$. The wave equation dataset describes acoustic propagation according to:

$$\partial_{\tau\tau} u = c^2 \Delta u, \quad u(\cdot, 0) = u_0, \quad \partial_{\tau} u(\cdot, 0) = v_0,$$

equipped with periodic boundary conditions. The wave speed is held constant at $c = 1$, the initial displacement u_0 is generated from a smooth periodic GRF, and the system initializes from rest ($v_0 = 0$). The exact continuous solution is derived using Fourier spectral methods and evaluated at 11 temporal intervals across $[0, 1]$. To integrate the endpoint Hermite residual, the governing dynamics are recast into a two-channel, first-order system with the state vector $q = (u, v)$:

$$F(q) = \begin{pmatrix} v \\ c^2 \Delta u \end{pmatrix}.$$

This first-order formulation drives the endpoint Hermite approximation, accompanied by periodic boundary enforcement.

Advection-diffusion equation. Let $(\mathbf{a}, \mathbf{u}) \triangleq (u_0, u_{\mathcal{T}})$. The advection-diffusion dataset models the interplay of convective and diffusive transport:

$$\partial_{\tau} u + b_x \partial_x u + b_y \partial_y u = \kappa \Delta u, \quad (x, y) \in [0, 1]^2,$$

constrained by periodic spatial boundaries. The initial state u_0 is instantiated from a smooth periodic GRF. The governing transport coefficients are independently drawn from uniform distributions: $b_x, b_y \sim \mathcal{U}(-1, 1)$ and $\kappa \sim \mathcal{U}(5 \times 10^{-4}, 5 \times 10^{-3})$. The analytical solution is resolved spectrally across 11 temporal frames up to $\mathcal{T} = 1$. The transport parameters (b_x, b_y, κ) are maintained as persistent scalar properties of the system. The endpoint Hermite residual is driven by:

$$F(u) = -b_x \partial_x u - b_y \partial_y u + \kappa \Delta u,$$

with rigorous periodic boundary enforcement.

Steady nonlinear heat conduction. Let $(\mathbf{a}, \mathbf{u}) \triangleq (f, u)$. This dataset investigates the nonlinear elliptic steady heat conduction problem defined by:

$$-\nabla \cdot (\lambda(u) \nabla u) = f(x, y), \quad \lambda(u) = \max\{1 + 0.05(u - 298), 0.1\}.$$

The system features mixed spatial boundary conditions:

$$u = u_D \text{ on the bottom boundary, } \quad \partial_{\mathbf{n}} u = 0 \text{ on the top, left, and right boundaries.}$$

The underlying Dirichlet boundary temperature is sampled uniformly as $u_D \sim \mathcal{U}(288, 308)$. The external forcing term f aggregates one to four Gaussian heat sources, each possessing an amplitude within $[10, 60]$, a standard deviation within $[0.04, 0.12]$, and a randomly placed interior centroid. The resulting nonlinear governing equation is resolved via Picard iteration, converging to a relative tolerance of 10^{-5} within a maximum of 30 iterations. The boundary temperature u_D acts as a preserved scalar metadata variable. The interior physical residual evaluates:

$$r_\Omega = -\nabla \cdot (\lambda(u) \nabla u) - f,$$

complemented by the required bottom Dirichlet and adjacent Neumann boundary residuals.

All model inputs are standardized channelwise using statistics computed from the training split:

$$z_c = \frac{x_c - \mu_c}{\sigma_c}, \quad \mu_c = \frac{1}{N_{\text{train}}HW} \sum_{n,i,j} x_{n,c,i,j}, \quad \sigma_c^2 = \frac{1}{N_{\text{train}}HW} \sum_{n,i,j} (x_{n,c,i,j} - \mu_c)^2.$$

The saved training normalizer is reused at test time.

4.2 Baseline Methods

Fourier Neural Operator (FNO). FNO (Li et al., 2021) serves as a supervised, full-grid neural operator baseline for forward modeling. It parameterizes the solution operator using spectral convolutions and is evaluated exclusively on complete input and output fields. We exclude vanilla FNO from the sparse-sensor experiments, as it lacks a native interface for sparse observations; similarly, inverse FNO is considered an adapted supervised operator and is omitted from the main comparative tables.

DeepONet. DeepONet (Lu et al., 2021) is a supervised operator-learning baseline comprising a branch network for input functions and a trunk network for query locations. We evaluate it on full forward maps, where the complete PDE condition is provided as input and the full solution field serves as the target. DeepONet variants adapted for sparse sensors or inverse problems are considered target-specific modifications and are excluded from the primary comparisons.

Invertible FNO (iFNO). iFNO (Long et al., 2025) acts as the full-grid invertible operator baseline. Because its architecture is explicitly designed for both forward and inverse operator learning, it is evaluated in both full forward and full inverse settings. Where the official implementation is unavailable, we utilize an architecture closely aligned with the published invertible FNO design.

RecFNO. RecFNO (Zhao et al., 2024) is employed as a baseline for amortized, sparse-sensor field reconstruction. It embeds sparse observations via mask- or Voronoi-based representations before reconstructing the global field using an FNO-style operator. Consequently, RecFNO is evaluated strictly on sparse solution reconstruction tasks, rather than on full-grid operator learning or sparse inverse problems.

Senseiver. Senseiver (Santos et al., 2023) is a sparse and irregular sensor reconstruction baseline utilizing an encoder-decoder architecture with coordinate queries. We apply it to

amortized sparse solution tasks in static or endpoint scenarios. It is also evaluated in time-varying data-assimilation settings when both trajectory query coordinates and time-varying sparse observations are available.

VoronoiCNN. VoronoiCNN (Fukami et al., 2021) projects sparse sensor measurements onto a Voronoi-tessellated grid and subsequently applies a convolutional neural network to reconstruct the complete field. It serves as an amortized sparse reconstruction baseline. Our implementation strictly follows the published CNN-Voronoi architecture; variants adapted for sparse inverse problems are omitted from the main tables.

PINN-Sparse. PINN-Sparse (Raissi et al., 2019) represents a per-instance, physics-constrained baseline for sparse observations. For each test sample, it optimizes a neural field by simultaneously minimizing a sparse observation loss and a structured physics loss, which incorporates interior PDE residuals alongside boundary and initial condition penalties. For sparse inverse problems, its application is restricted to static PDEs that possess an explicit coefficient/source or solution residual.

PC-BNN. PC-BNN (Sun and Wang, 2020) is a physics-constrained Bayesian neural network baseline updated via particle-based posterior methods. Within our primary comparisons, its native architectural assumptions align exclusively with the two-dimensional, three-channel shallow-water reconstruction setting. Consequently, it is evaluated solely on the Shallow-Water physics group for sparse solution reconstruction.

PDE-Constrained Optimization (PDE-Opt). PDE-Opt (Hinze et al., 2008) serves as a canonical per-instance optimization baseline. Rather than training an amortized predictor, it directly optimizes the unknown state or coefficient/source field for each individual sample, penalizing deviations from sparse observations, physical constraints, and smoothness priors. It is evaluated in both sparse solution reconstruction and static sparse inverse problems.

4D-Var. 4D-Var (Le Dimet and Talagrand, 1986) is the canonical variational data-assimilation baseline for time-dependent systems. It optimizes a complete state trajectory by balancing observation mismatches, background-state deviations, and model-dynamics residuals. We evaluate 4D-Var exclusively in the time-varying data-assimilation setting, utilizing loaded multi-timestep trajectories and time-varying sensor data.

VIVID. VIVID (Cheng et al., 2024) integrates an inverse-observation initialization module with variational trajectory refinement. In our primary evaluation, the inverse observation component is implemented to align with the official methodology, followed by subsequent observation, background, and dynamics refinement. Like 4D-Var, VIVID is evaluated strictly within the context of time-varying trajectory data assimilation.

4.3 Evaluation Settings

Our evaluation benchmark encompasses five representative Partial Differential Equation (PDE) families: Poisson, Helmholtz, Darcy, Burgers, and the Navier–Stokes equations. The experimental setup comprises 50,000 training samples, 1,000 validation samples, and 1,000 test samples. To systematically assess diverse learning paradigms, the benchmark is organized into seven distinct task groups.

Let $\mathbf{a}$ denote the PDE parameter or latent field (e.g., coefficient fields, source terms, or initial conditions), and let $\mathbf{u}$ denote the corresponding solution or terminal state. For time-dependent systems, $\mathbf{u}$ corresponds to the state at the target timestamp. Furthermore, let $\mathcal{O}_{\mathbf{a}}$ and $\mathcal{O}_{\mathbf{u}}$ denote the sparse observations of the PDE condition and solution field, respectively. The seven task groups are defined as follows:

Full Forward Problem. The full forward problem evaluates supervised, full-grid operator learning, mapping the complete PDE condition to the full solution domain:

$$\mathbf{a} \mapsto \mathbf{u}.$$

The participating baselines include FNO, DeepONet, and iFNO, evaluated across all five PDE families.

Full Inverse Problem. The full inverse problem considers full-grid inverse operator learning, mapping the complete solution field back to its underlying PDE parameter field:

$$\mathbf{u} \mapsto \mathbf{a}.$$

In accordance with our core benchmark configurations, this task is evaluated using iFNO across all five PDE families.

Sparse Solution Reconstruction. The sparse solution reconstruction task evaluates the recovery of a complete solution field from sparse spatial observations, denoted as:

$$\mathcal{O}_{\mathbf{u}} \mapsto \mathbf{u}.$$

This task encompasses two distinct paradigms depending on the inference mechanism:

- **Amortized Reconstruction:** Evaluates data-driven models pre-trained on the training set and directly deployed for fast inference on test samples. Participating baselines include RecFNO, Senseiver, and VoronoiCNN, evaluated across all five PDE families.
- **Physics-Based Reconstruction:** Assesses per-instance, physics-constrained optimization without prior training. The participating baselines are PINN-Sparse and PDE-Opt, optimized instance-by-instance across all five PDE families.

Both sub-tasks are benchmarked using 500 randomly placed, noise-free sensors per sample.

Sparse Forward Problem. The sparse forward problem addresses the estimation of the full solution domain given sparse spatial observations of the PDE input/condition:

$$\mathcal{O}_{\mathbf{a}} \mapsto \mathbf{u}.$$

This setting is evaluated under two operational paradigms:

- **Amortized Forward Prediction:** Evaluates data-driven baselines (RecFNO, Senseiver, and VoronoiCNN) pre-trained to directly predict the solution field from input observations across all five PDE families.
- **Physics-Based Forward Reconstruction:** Evaluates per-instance physics optimization using PINN-Sparse (budgeted at 1,000 optimization steps) and PDE-Opt (budgeted at 500 steps), restricted to Poisson, Helmholtz, and Darcy PDEs.

Both sub-tasks use 500 randomly placed, noise-free sensors per sample.

Sparse Inverse Problem. The sparse inverse problem evaluates parameter recovery from sparse observations of the solution:

$$\mathcal{O}_{\mathbf{u}} \mapsto \mathbf{a}.$$

This benchmark also incorporates both amortized and physics-based paradigms. The participating baselines comprise PINN-Sparse, PDE-Opt across Poisson, Helmholtz, and Darcy PDEs, and RecFNO, Senseiver, and VoronoiCNN, evaluated across all five PDE families with 500 randomly placed sensors.

Table 1: Summary of main evaluation configurations. “All PDEs” denotes the suite of five PDE families included in the main benchmark.

Task Setting	PDE Suite	Baseline Models
Full-grid forward	All PDEs	FNO, DeepONet, iFNO
Full-grid inverse	All PDEs	iFNO
Sparse solution reconstruction (amortized)	All PDEs	RecFNO, Senseiver, VoronoiCNN
Sparse solution reconstruction (physics-based)	All PDEs	PINN-Sparse, PDE-Opt
Sparse inverse (amortized)	All PDEs	RecFNO, Senseiver, VoronoiCNN
Sparse inverse (physics-based)	Poisson,Helmholtz,Darcy	PINN-Sparse, PDE-Opt
Sparse forward (amortized)	All PDEs	RecFNO, Senseiver, VoronoiCNN
Sparse forward (physics-based)	Poisson,Helmholtz,Darcy	PINN-Sparse, PDE-Opt

4.4 Results of Experiments

Quantitative results will be inserted after the experimental tables are finalized.

5 Discussion

5.1 Settings of Ablation Experiments

To ensure the robustness of the FM4PDE framework and to isolate the contributions of specific algorithmic components, we specify the following ablation protocol rather than treating the still-incomplete ablation results as established empirical findings. The baseline configuration across these ablations assumes simultaneous forward and inverse recovery tasks, stochastic sampling, endpoint-state loss evaluation, 500 noiseless observations, 100 integration steps, and a per-sample random sensor placement topology. PDE-specific baseline configurations dictate the dataset paths, channel semantics, boundary conditions, residual modes, and guidance weights. To maintain backward compatibility and ensure the reproducibility of previous model iterations, a legacy profile is retained in the codebase but is deliberately excluded from the formal default settings. The diagnostic ablations are structured to rigorously evaluate the following factors:

- **Guidance Components:** Evaluates whether recovery is primarily driven by the learned prior, observation guidance, PDE guidance, or a combination thereof. By comparing scenarios with no guidance, observation-only guidance, PDE-only guidance, and isolated coefficient/solution guidance, this study directly addresses the hypothesis that PDE guidance might be redundant if observation guidance dominates the objective.

- **Loss Evaluation State:** Tests the stability of finite-difference PDE residuals when evaluated on noisy intermediate flow states, one-step predicted states, or fully denoised endpoint estimates.
- **Sampler Phase:** Quantifies the practical effects of deterministic, stochastic, and hybrid guided trajectories. We evaluate various transition ratios between deterministic and stochastic phases to ascertain whether a deterministic warm start enhances or degrades the final reconstruction fidelity.
- **Guidance Schedule:** Investigates the necessity of time-adaptive guidance. By comparing constant, delta, linear, cosine, polynomial, and observation-decay schedules, this ablation determines whether constant guidance introduces bias or numerical instability near the terminal flow time.
- **Guidance Clipping:** Measures model stability and reconstruction accuracy as a function of gradient clipping. We compare theoretically unclipped updates against global-norm clipping (across thresholds from 1 to 10^{10}) and per-component clipping, documenting the divergence between practical implementation and theoretical sampling.
- **PDE Residual Region:** Isolates the effect of enforcing the PDE constraint globally versus localized enforcement (e.g., strictly within the interior boundary, or exclusively near observed locations).
- **Boundary-Condition Weighting:** Tests the system’s sensitivity to the explicit boundary condition residual strength ($\lambda_{bc} \in \{0.1, 1, 10\}$). This verifies that the true boundary enforcement mechanism is robust and analytically distinct from naive legacy boundary-zeroing behaviors.
- **Temporal Residual Mode:** Compares the coarse two-endpoint finite-difference (secant) residual against the more refined cubic Hermite bridge residual, thereby explicitly contextualizing endpoint-only temporal guidance as an approximation.
- **Near-Endpoint Temporal Residual:** Evaluates whether injecting sparse temporal observations near the initial and terminal boundaries improves the temporal derivative estimation compared to strictly endpoint-only guidance.
- **Sensor Sparsity:** Systematically reduces the observation count to track the degradation trajectories of the reconstruction error, observation loss, and PDE residual under genuinely sparse conditions. We consider $n_{\text{obs}} \in \{50, 100, 250, 500, 1000\}$.
- **Sensor Placement Topology:** Tests the model’s sensitivity to spatial observation distributions by comparing random, fixed, grid-based, and column-based sensor masks. This ensures that the framework’s efficacy is not artificially tied to a single, potentially biased mask topology.
- **Noise Robustness:** Assesses the utility and stability of the guidance mechanism when subjected to corrupted measurements, testing observation noise magnitudes of 0, 0.01, 0.05, and 0.10.

- **Time Grid and Integration Steps:** Generates error-versus-computational-cost curves by varying the temporal discretization (uniform or geometric grids), step counts (20, 50, 100, 200), and the numerical solver (Euler or midpoint). This isolates genuine algorithmic accuracy gains from brute-force step increases.
- **Task Flexibility:** Verifies the versatility of the learned joint prior by deploying it across distinct generative tasks: forward recovery, inverse recovery, simultaneous completion, and unconditional generation (establishing a zero-guidance baseline).
- **Guidance-Scale Sensitivity:** Quantifies the fundamental trade-off between PDE fidelity and observation matching by sweeping the PDE guidance scale, effectively establishing the observation-only scenario as a lower empirical bound. We consider $\zeta_{\text{pde}} \in \{0, 0.1, 1, 10, 100\}$.
- **Architecture Profile:** Applies light, base, and heavy network configurations across selected PDEs (e.g., Poisson, Helmholtz, Burgers, Navier-Stokes) to confirm that the observed methodological benefits generalize across varying model capacities.
- **Statistical Confidence:** Establishes rigorous mean and standard deviation bounds by aggregating reconstruction results across multiple orthogonal sources of randomness: stochastic sample seeds, observation mask seeds, and noise generation seeds.

5.2 Results of Ablation Experiments

Quantitative ablation results will be inserted after the experimental tables are finalized.

6 Conclusion

Acknowledgments and Disclosure of Funding

This work is supported by the National Key R&D Program of China 2025YFA1018700, the Beijing Outstanding Young Scientist Program (No.JWZQ20240101027), and the Beijing Natural Science Foundation (No. JR25003).

Appendix A. Definitions of Gaussian Random Field

This section specifies the Gaussian random fields (GRFs) used to generate coefficients, source terms, and initial conditions in the FM4PDE datasets. A GRF is a random function whose finite-dimensional marginals are jointly Gaussian. In the data generators below, GRFs are implemented through truncated spectral expansions on the numerical grid. The zero-frequency mode is removed, or equivalently each realization is centered, when a zero-mean field is required.

A.1 Static elliptic datasets: Darcy, Poisson, and Helmholtz

For the static elliptic datasets on $[0, 1]^2$, the MATLAB generator uses a zero-mean GRF with covariance operator

$$C_{\alpha, \tau} = (-\Delta_N + \tau^2 I)^{-\alpha},$$

where Δ_N denotes the Laplacian with homogeneous Neumann boundary conditions. On an $S \times S$ grid, the field is sampled by a cosine-basis Karhunen-Loeve expansion

$$G(x, y) = \sum_{m, n \geq 0}^{S-1} \lambda_{mn}^{1/2} \xi_{mn} \varphi_m(x) \varphi_n(y), \quad \xi_{mn} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1),$$

with the constant mode removed and

$$\lambda_{mn}^{1/2} \propto \tau^{\alpha-1} (\pi^2(m^2 + n^2) + \tau^2)^{-\alpha/2}.$$

The formal datasets use $\alpha = 3$ and $\tau = 4$ for these GRF draws. For the Darcy dataset, the generated Gaussian field is subsequently thresholded to form the binary coefficient

$$a(x, y) = \begin{cases} 12, & G(x, y) \geq 0, \\ 4, & G(x, y) < 0, \end{cases}$$

whereas Poisson and Helmholtz use the GRF directly as the source field.

A.2 One-dimensional Burgers dataset

For Burgers' equation on $[0, 1]$ with periodic boundary conditions, the initial condition is sampled from

$$u_0 \sim \mathcal{N}(m, C), \quad C = \sigma^2(-\Delta + \tau^2 I)^{-\gamma},$$

using a Fourier expansion. In the periodic case, the square-root eigenvalue for mode n has the form

$$\lambda_n^{1/2} = \sqrt{2} |\sigma| ((2\pi n)^2 + \tau^2)^{-\gamma/2}.$$

The dataset generator uses $m = 0$, $\gamma = 2.5$, $\tau = 7$, and $\sigma = 7^2$.

A.3 Periodic endpoint-pair datasets: Heat, Wave, and Advection-Diffusion

For the Python endpoint-pair generators on the periodic unit square, the initial field is obtained by filtering white noise in the Fourier domain. Let $W_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ and let $\widehat{W}(\boldsymbol{\kappa})$ denote its discrete Fourier transform. The filtered field is

$$\widehat{G}(\boldsymbol{\kappa}) = (\tau^2 + \|\boldsymbol{\kappa}\|_2^2)^{-s/4} \widehat{W}(\boldsymbol{\kappa}), \quad \widehat{G}(\mathbf{0}) = 0,$$

followed by inverse FFT and per-realization standardization,

$$\tilde{G} = \frac{G - \overline{G}}{\text{sd}(G) + \varepsilon}.$$

The generator then uses $\text{scale} \cdot \tilde{G}$ as the initial field. The default choices are summarized in Table 2.

Table 2: GRF parameters used by the periodic Python endpoint-pair generators.

Dataset	Field	Smoothness s	(τ, scale)
Heat	u_0	3.2	(5.0, 1.0)
Wave	u_0	5.0	(3.0, 0.5)
Wave, optional	v_0	5.5	(3.5, 0.1)
Advection-Diffusion	u_0	3.2	(5.0, 1.0)

A.4 Non-bounded Navier-Stokes dataset

For the periodic vorticity-form Navier-Stokes dataset, the initial vorticity is sampled by a Fourier GRF. For spatial dimension $d = 2$ and grid size S , the Fourier coefficient is generated as

$$\hat{\omega}_0(\mathbf{k}) = S^d \sqrt{2} \sigma (4\pi^2 \|\mathbf{k}\|_2^2 + \tau^2)^{-\alpha/2} (\xi_{\mathbf{k}} + i\eta_{\mathbf{k}}),$$

where $\xi_{\mathbf{k}}$ and $\eta_{\mathbf{k}}$ are independent standard normal variables and the zero mode is set to zero. If σ is not specified, the generator uses

$$\sigma = \tau^{(2\alpha-d)/2}.$$

The train/test GRF parameters are different: the training split uses $(\alpha, \tau) = (2.5, 7.0)$, whereas the test split uses $(\alpha, \tau) = (3.0, 6.5)$.

A.5 Reaction-Diffusion dataset

For the reaction-diffusion system, the spatial domain is

$$\Omega_{\text{RD}} = [-1, 1]^2,$$

and the simulator imposes homogeneous Neumann boundary conditions on the PDE solution. The formal FM4PDE data-generation wrapper supports both an i.i.d. standard-normal initial condition and a GRF initial condition. In the GRF mode, the two initial components are sampled independently from the same GRF family, then standardized and shifted/scaled:

$$u_0(x, y) = m_0 + \sigma_0 \tilde{G}_u(x, y), \quad v_0(x, y) = m_0 + \sigma_0 \tilde{G}_v(x, y),$$

where G_u and G_v are independent copies. The default values are

$$m_0 = 0, \quad \sigma_0 = 1, \quad \ell = 0.15, \quad p = 2.0,$$

and per-realization normalization is enabled.

A mathematically explicit form consistent with these generator parameters and with the Neumann PDE boundary is the Neumann spectral GRF

$$G_s(x, y) = \sum_{(m,n) \in \mathcal{K}} (1 + \ell^2 \lambda_{mn})^{-p/2} \xi_{mn}^{(s)} \varphi_m(x) \varphi_n(y), \quad s \in \{u, v\},$$

where $\xi_{mn}^{(s)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $\mathcal{K} = \{0, \dots, S-1\}^2 \setminus \{(0, 0)\}$, and

$$\lambda_{mn} = \left(\frac{m\pi}{2}\right)^2 + \left(\frac{n\pi}{2}\right)^2$$

are the eigenvalues of $-\Delta_N$ on $[-1, 1]^2$. The one-dimensional Neumann basis functions may be taken as

$$\varphi_0(x) = 1, \quad \varphi_m(x) = \sqrt{2} \cos\left(\frac{m\pi(x+1)}{2}\right), \quad m \geq 1,$$

up to the usual normalization constants on $[-1, 1]$. Equivalently, the covariance operator can be written as

$$C_{\ell, p}^{\text{RD}} = (I - \ell^2 \Delta_N)^{-p}.$$

Thus the parameter ℓ controls the spatial correlation length and p controls the high-frequency spectral decay. After sampling G_s , the realized field is standardized by

$$\tilde{G}_s(x_i, y_j) = \frac{G_s(x_i, y_j) - \bar{G}_s}{\sqrt{S^{-2} \sum_{i,j} (G_s(x_i, y_j) - \bar{G}_s)^2 + \varepsilon}}, \quad s \in \{u, v\},$$

with a small numerical $\varepsilon > 0$. This standardization means that the stored initial states are standardized GRF realizations: they retain the spatial smoothness implied by (A.5), but their empirical mean and variance are fixed sample by sample.

If the i.i.d. initialization mode is used instead, the initial values are not a spatial GRF; they are independent draws

$$u_0(x_i, y_j), v_0(x_i, y_j) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(m_0, \sigma_0^2).$$

The reported FM4PDE reaction-diffusion setting should therefore explicitly state which initialization mode is used and should not mix the GRF and i.i.d. modes in the same train/test split unless that mixture is part of the intended data distribution.

Appendix B. Flow Matching Network Architecture

All PDEs use a modified guided-diffusion-style U-Net as the flow-matching velocity network $v_\theta(\mathbf{x}_t, t)$. The model predicts a velocity field with the same data channel count as its input state. Physical scalar parameters such as α , c , b_x , b_y , κ , u_D , T , and Δt are stored as metadata for residual evaluation; they are not concatenated as constant image channels in the formal configuration. The model profile is selected by the PDE family. Poisson uses a smaller smooth elliptic profile; Darcy, Helmholtz, and Steady Heat Conduction use

a static elliptic profile; Heat, Advection-Diffusion, and Reaction-Diffusion use a temporal-endpoint base profile; Wave, Shallow-Water, and Navier-Stokes use a heavier temporal-endpoint profile; and Burgers uses a full time-space profile.

The U-Net contains a learned input projection, sinusoidal flow-time embedding followed by a two-layer MLP, residual blocks with scale-shift normalization, attention at selected downsampling factors, a residual-attention-residual bottleneck, and a skip-connected decoder. Optional value Fourier features lift the input field values through $\sin(\omega x)$ and $\cos(\omega x)$ channels; optional coordinate Fourier features lift deterministic grid coordinates. Attention is deliberately placed at coarser downsampling factors such as 8 and 16 rather than at factor 2, since attention cost is quadratic in the number of spatial tokens.

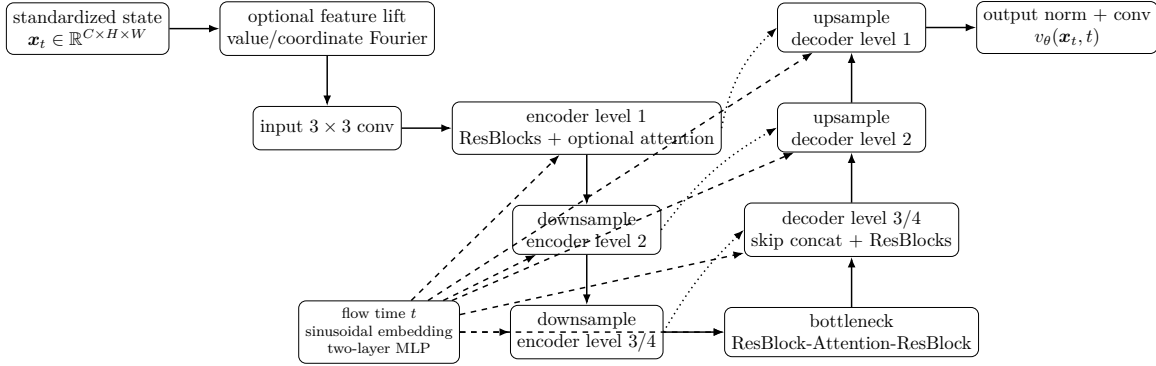


Figure 1: FM4PDE velocity-network schematic. The same U-Net template is used for all PDEs. Dashed arrows denote flow-time conditioning; dotted arrows denote encoder-decoder skip connections.

The recommended velocity-network configurations vary by PDE family to accommodate different physical complexities. The raw data channels and the effective input channels (which include internal Fourier feature channels added inside the U-Net) depend strictly on the state variables and coordinate dimensions inherent to each specific PDE.

The majority of the models—specifically those for Darcy, Helmholtz, Burgers, Reaction-Diffusion, Heat, Advection-Diffusion, and Steady Heat Conduction—employ a base network width of 128 with 4 residual blocks and channel multipliers of (1, 2, 4). Within this group, the static elliptic models (Darcy, Helmholtz, Steady Heat Conduction) utilize attention mechanisms at both $8\times$ and $16\times$ downsampled resolutions, whereas the others apply attention only at the $16\times$ scale. Fourier feature lifting is applied selectively based on the problem: Burgers and Helmholtz utilize both value and coordinate Fourier features; Darcy, Advection-Diffusion, and Steady Heat Conduction use only coordinate features; and Reaction-Diffusion and Heat rely solely on the raw data channels without additional Fourier lifting.

For more complex dynamics, such as Navier-Stokes, Shallow-Water, and Wave, a heavier temporal-endpoint architecture is deployed. These models feature an increased base width of 192 and deeper channel multipliers of (1, 2, 4, 4), with attention applied at downsampling factors of 8 and 16. Among these heavy models, Wave incorporates both value and coordi-

nate Fourier features, Navier-Stokes uses only value features, and Shallow-Water operates without them. Conversely, the Poisson equation uses a much lighter, smooth architecture designed with a base width of 96, 3 residual blocks, channel multipliers of (1, 2, 4), and attention only at the $16\times$ resolution, omitting all Fourier features.

The raw U-Net parameter counts for each PDE configuration are summarized in Table 3. These counts exclude optimizer states, exponential moving average (EMA) shadow weights, and checkpoint metadata.

Table 3: Channel definitions, variable conventions, and parameter counts for the recommended FM4PDE velocity networks.

PDE	Channel Definition	Exact Parameters
Poisson	$[f, \phi]$	44,121,218
Heat	$[u_0, u_T]$	96,434,306
Reaction-Diffusion	$[u_0, v_0, u_T, v_T]$	96,438,916
Darcy	$[a, p]$	96,455,042
Advection-Diffusion	$[u_0, u_T]$	96,455,042
Steady Heat Conduction	$[f, u]$	96,455,042
Burgers	$[u]$ (as a time-space image)	96,457,345
Helmholtz	$[f, \psi]$	96,464,258
Navier-Stokes	$[\omega_0, \omega_T]$	387,487,682
Shallow-Water	$[h_0, hu_0, hv_0, h_T, hu_T, hv_T]$	387,487,686
Wave	$[u_0, v_0, u_T, v_T]$	387,539,524

Appendix C. Training Details

The training process minimizes the conditional optimal-transport flow-matching objective:

$$\min_{\theta} \mathbb{E}_{X_1, X_0, t} \|v_{\theta}(X_t, t) - v_t(X_t \mid X_1)\|_2^2, \quad (108)$$

where X_1 represents a standardized PDE data pair, X_0 is standard Gaussian noise of the identical tensor shape, and X_t is sampled from the conditional probability path. Unless skewed timestep sampling is explicitly necessitated, the temporal variable is uniformly sampled as $t \sim \mathcal{U}(10^{-5}, 1 - 10^{-5})$.

The standard training protocol employs the following hyperparameters and configurations:

- **Optimization.** The network is optimized using AdamW with a base learning rate of 10^{-4} , momentum parameters $\beta = (0.9, 0.999)$, numerical stability term $\epsilon = 10^{-8}$, and a decoupled weight decay factor of 0.01.
- **Learning-Rate Schedule.** Training utilizes a linear warmup and cosine annealing schedule. The learning rate is linearly increased from 10^{-5} to 10^{-4} over the initial 5 epochs, followed by a cosine decay down to a minimum of 10^{-6} .

- **Epochs and Batch Strategy.** The models are trained for a total of 300 epochs. To maintain optimization stability across varying computational constraints, the target effective batch size is strictly maintained at 64. For most PDEs, this is achieved using a per-GPU batch size of 4; however, for memory-intensive families (e.g., Reaction-Diffusion, Shallow-Water, and Wave), the batch size is reduced to 2, supplemented by gradient accumulation to ensure the effective batch size requirement is met.
- **Data Loading and Scale.** Pretraining leverages large-scale simulation data, typically corresponding to 50,000 paired training samples per PDE dataset. Subsequent statistical evaluations are rigorously conducted on an independent test set of 10,000 samples to ensure robust performance estimation.
- **Spatial and Temporal Resolution.** The default computational grid is discretized to 128×128 . For the Burgers’ equation, this represents a grid of 128 physical-time intervals by 128 spatial points; for all other static or temporal-endpoint PDEs, it denotes a dense two-dimensional spatial domain.
- **Standardization.** Data normalization is strictly derived from the training set to prevent data leakage. Channel-wise means and standard deviations are computed globally over the training tensors and frozen into the checkpoint metadata. These fixed statistics are subsequently utilized for inverse transformation during the generation and inference phases.
- **Precision and Evaluation.** The models are natively trained in standard 32-bit floating-point precision. Periodic model checkpoints and lightweight residual diagnostics on generated samples are logged every 50 epochs to track convergence and prevent overfitting.

Appendix D. Guidance Hyperparameters

The default sparse-sampling weights used by the base configs are detailed below. Here $\zeta_{\mathbf{a}}$ is the coefficient/source/initial-side observation weight, $\zeta_{\mathbf{u}}$ is the solution/terminal-side observation weight, and ζ_{pde} is the PDE residual weight. Unless varied by ablation, the stochastic guidance coefficient is $c_{\zeta} = 0.1$, the loss state is endpoint, the gradient target is `current_state_chain_rule`, global-norm clipping is enabled with threshold 10^{10} , the time grid is uniform, the number of steps is 100, and the numerical stepper is Euler.

References

- G. Baldan, Q. Liu, A. Guardone, and N. Thuerey. Flow matching meets PDEs: A unified framework for physics-constrained generation. *arXiv preprint arXiv:2506.08604*, 2025.
- P. G. Bissiri, C. C. Holmes, and S. G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(5):1103–1130, 2016. doi: 10.1111/rssb.12158.
- S. Boucheron, G. Lugosi, and P. Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

Table 4: Default sparse-sampling guidance weights for each PDE.

PDE	$\zeta_{\mathbf{a}}$	$\zeta_{\mathbf{u}}$	ζ_{pde}
Darcy	1.0×10^5	5.0×10^7	1
Poisson	6.0×10^4	5.0×10^6	1
Helmholtz	2.0×10^5	2.0×10^7	1
Navier-Stokes	3.0×10^4	3.0×10^5	1
Burgers	-	3.2×10^3	10
Reaction-Diffusion	4.0×10^7	4.0×10^7	10
Shallow-Water	2.0×10^5	2.0×10^5	10
Heat	1	1	1
Wave	1	1	1
Advection-Diffusion	1	1	1
Steady Heat Conduction	1	1	1

- S. Cheng, C. Liu, Y. Guo, and R. Arcucci. Efficient deep data assimilation with sparse observations and time-varying sensors. *Journal of Computational Physics*, 496:112581, 2024. doi: 10.1016/j.jcp.2023.112581.
- K. Fukami, R. Maulik, N. Ramachandra, K. Fukagata, and K. Taira. Global field reconstruction from sparse sensors with Voronoi tessellation-assisted deep learning. *Nature Machine Intelligence*, 3(11):945–951, 2021. doi: 10.1038/s42256-021-00402-2.
- E. Hairer, S. P. Nørsett, and G. Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, 1993.
- M. Hinze, R. Pinnau, M. Ulbrich, and S. Ulbrich. *Optimization with PDE Constraints*, volume 23 of *Mathematical Modelling: Theory and Applications*. Springer, Dordrecht, 2008. doi: 10.1007/978-1-4020-8839-1.
- J. Ho and T. Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.
- J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020.
- J. Huang, G. Yang, Z. Wang, and J. J. Park. DiffusionPDE: Generative PDE-solving under partial observation. *Advances in Neural Information Processing Systems*, 37:130291–130323, 2024.
- C. Jacobsen, Y. Zhuang, and K. Duraisamy. CoCoGen: Physically consistent and conditioned score-based generative models for forward and inverse problems. *SIAM Journal on Scientific Computing*, 47(2):C399–C425, 2025. doi: 10.1137/24M1636071.
- H. Karimi, J. Nutini, and M. Schmidt. Linear convergence of gradient and proximal-gradient methods under the Polyak-Łojasiewicz condition. In *Proceedings of the European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 795–811. Springer, 2016.

- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- F.-X. Le Dimet and O. Talagrand. Variational algorithms for analysis and assimilation of meteorological observations: Theoretical aspects. *Tellus A: Dynamic Meteorology and Oceanography*, 38A(2):97–110, 1986. doi: 10.1111/j.1600-0870.1986.tb00459.x.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=c8P9NQVtmn0>.
- Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=PqvMRDCJT9t>.
- Y. Lipman, M. Havasi, P. Holderrieth, N. Shaul, M. Le, B. Karrer, R. T. Q. Chen, D. Lopez-Paz, H. Ben-Hamu, and I. Gat. Flow matching guide and code. *arXiv preprint arXiv:2412.06264*, 2024.
- X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=XVjTT1nw5z>.
- D. Long, Z. Xu, Q. Yuan, Y. Yang, and S. Zhe. Invertible Fourier neural operators for tackling both forward and inverse problems. In Y. Li, S. Mandt, S. Agrawal, and E. Khan, editors, *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 3043–3051. PMLR, 2025. URL <https://proceedings.mlr.press/v258/long25a.html>.
- L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021. doi: 10.1038/s42256-021-00302-5.
- M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.
- J. E. Santos, Z. R. Fox, A. Mohan, D. O’Malley, H. Viswanathan, and N. Lubbers. Development of the Senseiver for efficient field reconstruction from sparse observations. *Nature Machine Intelligence*, 5:1317–1325, 2023. doi: 10.1038/s42256-023-00746-x.
- Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=PXTIG12RRHS>.

- L. Sun and J.-X. Wang. Physics-constrained bayesian neural network for fluid flow reconstruction with sparse and noisy data. *Theoretical and Applied Mechanics Letters*, 10(3): 161–169, 2020. doi: 10.1016/j.taml.2020.01.031.
- U. Utkarsh, P. Cai, A. Edelman, R. Gomez-Bombarelli, and C. V. Rackauckas. Physics-constrained flow matching: Sampling generative models with hard constraints. *arXiv preprint arXiv:2506.04171*, 2025.
- L. Yang, X. Meng, and G. E. Karniadakis. B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021. doi: 10.1016/j.jcp.2020.109913.
- L. Zhang, A. Rao, and M. Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3836–3847, 2023.
- X. Zhao, X. Chen, Z. Gong, W. Zhou, W. Yao, and Y. Zhang. RecFNO: A resolution-invariant flow and heat field reconstruction method from sparse observations via Fourier neural operator. *International Journal of Thermal Sciences*, 195:108619, 2024. doi: 10.1016/j.ijthermalsci.2023.108619.